CLASSIFICATION OF BIPARTITE OBSERVABLES III: SIMPLICITY, MIXTURES AND QUANTUM ERROR CORRECTION

EDUARDO DUEÑEZ AND JOSÉ IOVINO

Abstract. This is the third of a series of three papers that bring model theory to quantum information. The dividing lines of Shelah's classification theory, which separate the tame first-order theories from the wild ones, become properties of bipartite observables: properties of the expectation values obtained when two independently prepared quantum systems are idealized one after the other. In this paper we focus on simple unstable theories. The Pauli group of quantum error correction, with anticommutation as its relation, has a simple unstable theory, like the random graph. Forking and the Kim–Pillay independence theorem acquire a meaning in stabilizer codes: forking recognizes logical errors, and the independence theorem glues syndromes.

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1. Introduction

The paradigm of a simple unstable theory is the theory of the random graph [She80, GIL02]. A first-order theory, a set of axioms in a formal language, is simple, in the sense of Shelah, when no formula has the tree property, a branching pattern of consistent and inconsistent conditions that we describe later in this Introduction; the theory of the random graph is

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unstable because its edge relation has the independence property: for any finitely many vertices and any choice among them, some vertex is adjacent to the chosen ones and to none of the others. The random graph has two modes of presentation: as the Rado graph, it is the Fraïssé limit of the finite graphs, that is, the countable graph, unique up to isomorphism, that contains every finite graph as an induced subgraph and in which every isomorphism between finite induced subgraphs extends to an automorphism [Fra54, Rad64]; as a random object, it is the graph to which a random network of couplings on countably many sites, in which each pair of sites is coupled independently with probability one half, is isomorphic with probability one [ER63]. Quantum error correction, the protection of quantum information against noise, rests on a structure of its own. In a stabilizer code, errors are detected by measuring checks, which are Pauli strings: tensor products of factors on the individual qubits, or two-level quantum systems, each factor being the identity, a bit flip, a phase flip or, up to a phase, their product. The checks of a code commute, and the code stores quantum information in their common eigenspace for the eigenvalue 1. It suffices to correct errors that are themselves Pauli strings, since a code that corrects a set of errors corrects every linear combination of them (see [Got97]).

To make this precise, let $\mathbb{F}_2$ be the two-element field and let n be a positive integer. The Hilbert space of a qubit is $\mathbb{C}^2$, on which the bit flip X and the phase flip Z, two of the Pauli matrices, exchange the two vectors of the standard basis and change the sign of the second, respectively. A Pauli string on n qubits is a tensor product of n factors $X^{x_k}Z^{z_k}$, for $k = 1, \dots, n$, with $x_k, z_k \in \mathbb{F}_2$, multiplied by one of the phases ± 1 and $\pm \mathrm{i}$; up to its phase, it is determined by its label, the vector $(x, z) \in \mathbb{F}_2^{2n}$ with $x = (x_1, \dots, x_n)$ and $z = (z_1, \dots, z_n)$. Let ω be the form on $\mathbb{F}_2^{2n}$ given by $\omega((x, z), (x', z')) = x \cdot z' + z \cdot x'$ for $x, z, x', z' \in \mathbb{F}_2^n$, where $x \cdot z' = \sum_{k=1}^n x_k z'_k$; it is bilinear, alternating and nondegenerate, and it is the standard symplectic form on $\mathbb{F}_2^{2n}$. Two Pauli strings on n qubits either commute or anticommute, that is, their products in the two orders are equal or opposite, and their labels decide which: the strings anticommute exactly when ω takes the value 1 at their labels. We disregard phases and identify each Pauli string with its label. Measuring a check g on a state that g fixes, after an error e, itself a Pauli string on the n qubits, has acted on the state, reveals whether the two strings anticommute: the syndrome bit of e with respect to g is 1 if they anticommute and 0 if they commute, and in terms of labels it is $\omega(g, e)$. The syndrome of e with respect to checks $g_1, \dots, g_m$, for a positive integer m, is the vector of its syndrome bits with respect to $g_1, \dots, g_m$.

On countably many qubits, the labels of the Pauli strings that act nontrivially on only finitely many qubits form a vector space over $\mathbb{F}_2$ of countably infinite dimension, on which the form ω, given by the same formula, is nondegenerate. This countable symplectic space is, under our identification, the Pauli group on countably many qubits taken modulo phases, which we call simply the Pauli group. In Section 3 we show that, with this form, the Pauli group is the Fraïssé limit of the finite alternating spaces over $\mathbb{F}_2$, the finite vector spaces over $\mathbb{F}_2$ with an alternating bilinear form, and that its theory, like that of the random graph, is simple and unstable. For each of the two structures, we prove simplicity by exhibiting an explicit relation of independence between two sets over a third and verifying for it the Kim–Pillay axioms, which ensure that the theory is simple (see Fact 2.2). The Pauli group is thus the quantum counterpart of the random graph, and in this paper we study the two paradigms side by side.

In Section 4 we give a dictionary between model theory and quantum error correction, and we prove the three results on which it rests. The checks of a stabilizer code generate

its stabilizer group; an error is undetectable when it commutes with every check, and an undetectable error is a logical error when it alters the encoded information, that is, when it lies outside the stabilizer group. Clifford unitaries, the unitary operators that map Pauli strings to Pauli strings under conjugation, up to phases, implement the homogeneity of the Pauli group: every isomorphism between finite-dimensional substructures of the Pauli group extends to an automorphism induced by a Clifford unitary on finitely many qubits, and every stabilizer code has an encoding circuit made of Clifford gates. The type of an error over a set of Pauli strings records which elements of the set anticommute with it and how, if at all, it is a product of them. An extension of such a type to a larger set forks over the original set when it confines the error to one of infinitely many alternatives that the original type cannot tell apart. Forking recognizes the logical errors: an undetectable error is a logical error exactly when its type over the stabilizer group and a set of logical operators, which generate the undetectable errors modulo the stabilizer group, forks over the stabilizer group. The Kim–Pillay independence theorem becomes a rule for syndromes: if two sets of checks are independent over a common set, then any two syndromes, each realized by some error on its own set of checks, that agree on the common set are realized by a single error. Here two sets of checks are independent over a common set when a Pauli string that is, up to a phase, both a product of checks of the first set and the common set and a product of checks of the second set and the common set is already, up to a phase, a product of checks of the common set. As first-order structures the two paradigms are alike: both are Fraïssé limits, and both have simple unstable theories. We show that the bipartite observables that they define, which we introduce in the next two paragraphs, tell them apart.

This paper is the third of a series on bipartite observables. A bipartite observable is a bounded self-adjoint operator on the tensor product of the Hilbert spaces of two systems, one prepared by Alice and the other probed by Bob. A preparation of Alice, a state of her system, averages the observable to an operator on Bob's space. These operators and the identity generate the probe algebra, the smallest algebra of operators on Bob's space that contains them and is closed under adjoints and under limits in the operator norm. A probe of Bob is a state of the probe algebra, that is, a positive linear functional on it with value 1 at the identity; it assigns to the operator of each preparation a real number, the response of the preparation at the probe. In the first paper [DI26a] we studied stability in the sense of Shelah: an observable is stable when its expectation, for independently prepared systems, does not depend on which of the two systems is idealized first. A system is idealized when its state is replaced by the limit, along a nonprincipal ultrafilter on the natural numbers, of a sequence of its states given by density operators, the positive operators of trace one. This limit assigns to each bounded operator on the space of the system the ultralimit of its expectations in the states of the sequence, that is, the unique complex number that these expectations approximate, to within any prescribed error, on a set of indices belonging to the ultrafilter. In the second paper [DI26b] we called a sequence of Alice's preparations independent when there are real numbers $\alpha < \beta$ such that, for any two disjoint finite sets of these preparations, some probe gives responses at most α on the first set and at least β on the second; an observable with no independent sequence is NIP, an acronym for "no independence property". We showed there that an observable is stable exactly when its idealized responses, the responses of its idealized preparations, are continuous functions on the probe space, the set of probes with the weak* topology of pointwise convergence on the probe algebra, and that it is NIP exactly when they are of the first Baire class, that is, pointwise limits of sequences of continuous functions. The idealized responses of a NIP observable form a Rosenthal compactum, a compact set, in the topology of pointwise convergence, of functions of the first

Baire class on a compact metrizable space; this compactum is separable, and Todorčević's trichotomy for separable Rosenthal compacta applies to it. Simplicity is another of Shelah's dividing lines, the properties by which he classified first-order theories; it neither implies nor follows from NIP, and a theory is stable exactly when it is both simple and NIP. It is, moreover, a property of a different kind: stability and NIP of an observable are characterized by properties of the closure of its family of responses in the topology of pointwise convergence, whereas simplicity concerns the joint consistency of conditions on an unknown element, such as a probe. A theory fails to be simple exactly when, for some formula and some integer $k \geq 2$, there is a tree of instances of the formula, indexed by the finite sequences of natural numbers, whose branches are consistent while any k distinct children of each node are jointly inconsistent. In error correction, the unknown element is an error, and the conditions are statements about its syndrome.

In Section 5 we define the observables of the two paradigms. Both are diagonal observables: the Hilbert spaces of Alice and Bob have orthonormal bases indexed by countable sets of labels, and the observable reports whether a relation holds between Alice's label and Bob's. In the Rado observable, Alice and Bob each hold a site of the network, that is, a vertex of the Rado graph, and the observable reports whether the two sites are coupled; in the Pauli observable, Alice holds a check and Bob an error, and the observable reports whether the two anticommute. A mixed preparation of Alice, a density operator on her space, enters a diagonal observable only through its diagonal, a probability measure on her labels; a single mixed preparation can therefore spread its weight over arbitrarily many labels, and its response is affine in the mixture. We call the point masses at single labels her pure preparations. Bob's probes differ in the two observables, as we establish in Lemma 5.5 and Proposition 5.4: for the Rado observable they are the Radon probability measures on the space of all sets of vertices, with the product topology, that is, the random sets of neighbors, and for the Pauli observable they are the Pauli channels, the probability laws of random Pauli errors of arbitrary support, the support of an error being the set of qubits on which it acts nontrivially. The pure probes are the point masses among the probes, and we identify each of them with the point at which it is concentrated. Every set of vertices is thus a pure probe of the Rado observable; we express this by saying that the pure responses of this observable, the responses of pure preparations, are freely independent. The pure probes of the Pauli observable, its idealized errors, are the Pauli strings of arbitrary support. The syndrome sector of a syndrome with respect to finitely many checks is the set of idealized errors with that syndrome. The depolarizing noise of rate p, for $0 < p \leq 1$, which acts independently on each qubit, leaving it unchanged with probability $1 - p$ and applying a bit flip, a phase flip or their product, each with probability $p/3$, is a Pauli channel concentrated on the Pauli strings of infinite support, which lie outside the Pauli group; it models the noise of a quantum memory on countably many qubits.

The real-valued theory of an observable is the theory of the structure formed by its mixed preparations, its probes and its response, in Keisler's real-valued logic [Kei23], whose formulas take real values. A formula has the tree property of the second kind when, for some integer $k \geq 2$, there is an array of instances of the formula, with rows and columns indexed by the natural numbers, in which any k distinct instances of a row are jointly inconsistent while every path, which chooses one instance from each row, is consistent. In real-valued logic these conditions are inequalities with a gap: a sequence of instances is uniformly k-inconsistent when, for some reals $\alpha < \beta$, each instance takes a value at least β somewhere while no k of them exceed α at a common point. We prove in Section 6 that the real-valued theory of every diagonal observable with the independence property, and in particular of the Rado observable and of the Pauli observable, has the tree property of the second kind: the

difference between the responses to the uniform mixtures on a half of a finite set of labels and on the complementary half reaches its maximum, 1, at a probe exactly when the probe answers 1 on the first half and 0 on the second, and the differences given by two halves that meet in a quarter of the set are never both greater than 1/2. Actual preparations are those of the structure of the observable, and idealized preparations are the new preparations in its elementary extensions, the larger structures, such as its ultrapowers, in which every formula with parameters from the structure takes the same value as in the structure itself. In an ultrapower along a nonprincipal ultrafilter on the natural numbers, the preparations and the probes are given by sequences $(\mu_k)_{k\in\mathbb{N}}$ of mixed preparations and $(\nu_k)_{k\in\mathbb{N}}$ of probes, respectively; the response of the preparation at the probe is the limit, along the ultrafilter, of the response of μ_k at ν_k. In Sections 7 and 8 we show that, for the Pauli observable, the tree property of the second kind is witnessed by an array of actual preparations, while, for the Rado observable, every array that witnesses it for a quantifier-free formula contains idealized preparations. Finally, in Section 9 we compare mixing in our sense with the affine logic of Ben Yaacov, Ibarlucía and Tsankov, in which a mixture of structures is a direct integral, and we close with open questions in Section 10.

Theorem 1.1 (Main results).

(A) *The Rado graph and the symplectic space of countably infinite dimension over $\mathbb{F}_2$, which is the Pauli group on countably many qubits modulo phases, have simple unstable theories (Theorems 3.1 and 3.3).*

(B) *Let n be a positive integer, let a stabilizer code on n qubits be given, let S be the set of labels of its stabilizer group, and let L be a set of logical operators of the code, labels that together with S span the labels of its undetectable errors; labels on n qubits are regarded as elements of the Pauli group through the first n of countably many qubits. An undetectable error is a logical error exactly when its type over $S \cup L$, in the theory of the Pauli group, forks over S; all logical errors have the same type over S, and this type does not fork over the empty set (Theorem 4.5 and Proposition 4.7). Every isomorphism between finite-dimensional substructures of the Pauli group extends to an automorphism induced by a Clifford unitary on finitely many qubits, and every stabilizer code has an encoding circuit made of Clifford gates (Theorem 4.3 and Corollary 4.4). The independence theorem becomes the gluing of syndromes: if two sets of checks are independent over a common set, then any two syndromes, each realized by some error on its own set of checks, that agree on the common set are realized by a single error (Proposition 4.8). The probes of the Pauli observable are the Pauli channels, and the depolarizing noise of every positive rate on countably many qubits is concentrated on the Pauli strings of infinite support, which are idealized errors and not actual ones (Proposition 5.4).*

(C) *For every diagonal observable whose pure responses, the responses of single labels, have the independence property, and in particular for the Rado observable and the Pauli observable, the real-valued theory of the observable has the tree property of the second kind, for the formula that is the difference of two responses. Since the independence property of a diagonal observable passes to its pure responses, it follows that a diagonal observable whose real-valued theory is simple is NIP (Theorem 6.1).*

(D) *For the Pauli observable, the indicator of the syndrome sector of every nonzero syndrome of finitely many linearly independent checks is, on pure probes, the difference of the responses of two uniform mixtures of checks, and the response itself has the tree property of the second kind, witnessed by actual preparations: for blocks of checks that are linearly independent all together, the conditions that a given check is the first of its block to fire, that is, to anticommute with the unknown error, form the rows of a witnessing array; in the bit-flip repetition code, whose checks compare neighboring qubits of a chain, the first check to fire marks the first domain wall of a bit flip, the first boundary between flipped and unflipped qubits (Proposition 7.1, Theorem 7.2 and Example 7.4).*

(E) *For the Rado observable, and more generally for every diagonal observable with freely independent pure responses, that is, one in which every pattern of answers to Alice's labels is a pure probe, such as the observable of the random bipartite graph, the Fraïssé limit of the finite bipartite graphs with the two sides as two sorts, the conditions on a probe given by instances of a quantifier-free formula with actual mixtures as parameters are compatible along every sequence: no such sequence is uniformly k-inconsistent, for any integer $k \geq 2$. Hence, for these observables, every array that witnesses the tree property of the second kind of the formula in (C) contains idealized preparations (Theorems 8.1 and 8.3, and Corollary 8.2).*

The difference between (D) and (E) lies in the probes. For the Pauli observable we show in Proposition 5.4 that the pure probes are the Pauli strings of arbitrary support, which form a compact group in the product topology. Alice's pure responses are then, up to an affine change, the characters of this group, that is, its continuous homomorphisms into the multiplicative group $\{\pm 1\}$. Products of characters are characters, and the indicators of syndrome sectors can therefore be computed by averaging: for finitely many linearly independent checks and a nonzero syndrome, the indicator of its syndrome sector is, on pure probes, the average response over those elements of the group generated by the checks which anticommute with the errors of the sector, minus the average response over those which commute with them (see Proposition 7.1); this identity is the counterpart, for responses, of the familiar expression of the projector onto the states of a given syndrome as an average over the stabilizer group [Got97], although it requires the checks only to be linearly independent, not to commute. With it, a single mixture of checks can single out the first check of a block to fire, that is, to anticommute with a given error: for linearly independent checks $g_0, g_1, \dots$ and each $j \in \mathbb{N}$, the uniform mixture on the coset of g_j modulo the span of the earlier checks has response 1 at an error exactly when g_j is the first of these checks to anticommute with the error. For the Rado observable, by contrast, we show in Lemma 5.5 that the pure probes are all the sets of neighbors, which form the whole Cantor cube, the space of all subsets of the vertex set with the product topology, and that the pure responses are its coordinate functions. Averages of coordinates depend on a probe only through its marginals, the probabilities with which it declares the individual vertices neighbors; in the proof of Theorem 8.1, a gliding hump argument, which passes to a subsequence whose weights lie, up to small errors, on a common finite set and on pairwise disjoint finite blocks, shows that no sequence of conditions given by affine combinations of responses of actual mixtures, with a common bound on the sum of the absolute values of the constant term and the coefficients, is uniformly k-inconsistent, for any $k \geq 2$. The same gliding hump, in which maximizers of the individual conditions are glued on disjoint blocks of labels, extends this conclusion to the conditions given by quantifier-free formulas, which are continuous functions of finitely many responses (see Theorem 8.3). For the observable of the paradigm, the Rado graph, every array of mixtures that witnesses the tree property of the second kind for a quantifier-free formula therefore contains idealized preparations; for the observable of its quantum counterpart, the Pauli group, the linear relations among the labels, which make the syndrome bit of a product of checks the sum of their syndrome bits, allow actual preparations to witness the same property.

2. Simple theories and the Kim–Pillay theorem

In this section we recall the tree property, whose absence defines simple theories, together with its two variants, and the Kim–Pillay theorem, which characterizes simple theories by an independence relation; in Section 3 we use this theorem to prove that the theories of both paradigms, the Rado graph and the Pauli group, are simple.

To fix the setting, we work in a monster model $\mathbb{M}$ of a complete first-order theory T: a model of very large cardinality in which every set of formulas in finitely many variables with parameters from a subset of smaller cardinality is satisfied whenever it is finitely satisfiable, and in which every elementary map between two such subsets, that is, every map under which each tuple and its image satisfy the same formulas, extends to an automorphism. These subsets are called small, and from now on all sets of parameters are assumed to be small. The type $\mathrm{tp}(a/C)$ of a tuple a over a set C is the set of formulas with parameters

from C that a satisfies. For sets B and D we write BD for $B \cup D$, and, identifying a tuple a with the set of its entries, we write aB for the union of that set with B. For a finite sequence $s \in \mathbb{N}^{<\mathbb{N}}$ of natural numbers and $j \in \mathbb{N}$, we write $s^\frown j$ for the sequence obtained by appending j to s, and for an infinite sequence $\eta \in \mathbb{N}^{\mathbb{N}}$ and $n \in \mathbb{N}$, we write $\eta \restriction n$ for the sequence of the first n terms of η. Two finite sequences are incomparable when neither is an initial segment of the other.

Definition 2.1. Let $\varphi(x, y)$ be a formula, where x and y are finite tuples of variables.

(1) For an integer $k \geq 2$, a sequence $(b_j)_{j\in\mathbb{N}}$ of tuples of $\mathbb{M}$ of the length of y is *k-inconsistent for φ* if for all $j_1 < \cdots < j_k$ in $\mathbb{N}$ there is no x with $\varphi(x, b_{j_1}) \wedge \cdots \wedge \varphi(x, b_{j_k})$.
(2) The formula φ has the *tree property* if, for some $k \geq 2$, there is a tree $(b_s)_{s\in\mathbb{N}^{<\mathbb{N}}}$ such that for every $\eta \in \mathbb{N}^{\mathbb{N}}$ the set $\{\varphi(x, b_{\eta\restriction n}) : n \in \mathbb{N}\}$ is consistent and for every s the sequence $(b_{s^\frown j})_{j\in\mathbb{N}}$ is k-inconsistent for φ.
(3) The formula φ has the *tree property of the first kind* if there is a tree $(b_s)_{s\in\mathbb{N}^{<\mathbb{N}}}$ such that for every $\eta \in \mathbb{N}^{\mathbb{N}}$ the set $\{\varphi(x, b_{\eta\restriction n}) : n \in \mathbb{N}\}$ is consistent and for all incomparable $s, t \in \mathbb{N}^{<\mathbb{N}}$ the set $\{\varphi(x, b_s), \varphi(x, b_t)\}$ is inconsistent.
(4) The formula φ has the *tree property of the second kind* (TP_2) if, for some $k \geq 2$, there is an array $(b_{ij})_{i,j\in\mathbb{N}}$ such that every row $(b_{ij})_{j\in\mathbb{N}}$ is k-inconsistent for φ and for every $f\colon \mathbb{N} \to \mathbb{N}$ the set $\{\varphi(x, b_{if(i)}) : i \in \mathbb{N}\}$ is consistent.
(5) The theory T is *simple* if no formula has the tree property.

Recall that the theory T is stable when no formula $\varphi(x, y)$ has the order property, that is, when there are no tuples a_i and b_i, for $i \in \mathbb{N}$, such that, for all $i, j \in \mathbb{N}$, the formula $\varphi(a_i, b_j)$ holds exactly when $i < j$. It is NIP when no formula $\varphi(x, y)$ has the independence property, that is, when there are no tuples b_i, for $i \in \mathbb{N}$, such that every subset of $\mathbb{N}$ has the form $\{i \in \mathbb{N} : \varphi(a, b_i) \text{ holds}\}$ for some tuple a. Every stable theory is simple, and the tree property of the second kind implies the tree property; by a theorem of Shelah, a theory is simple exactly when it has neither TP_2 nor the tree property of the first kind (see [She90, Theorem III.7.11]). A simple theory is stable if and only if it is NIP (see [She80, GIL02]), so a simple unstable theory has the independence property. To prove simplicity we do not build trees; instead, we exhibit an independence relation directly and show that it is nonforking independence, which we now recall. For a formula $\varphi(x, y)$ and a tuple b of $\mathbb{M}$ of the length of y, the formula $\varphi(x, b)$ *divides* over a set C when, for some integer $k \geq 2$, there is a sequence $(b_j)_{j\in\mathbb{N}}$ with $b_0 = b$ that is k-inconsistent for φ and indiscernible over C, that is, such that any two of its increasing finite subsequences of the same length have the same type over C. The formula *forks* over C when it implies a finite disjunction of formulas each of which divides over C; a type forks over C when one of its formulas does. Nonforking independence is the relation under which $A \mathop{\smile\!\!\!\!\!\!|}_C B$ holds, for small sets A, B and C, when, for every finite tuple a from A, the type $\mathrm{tp}(a/BC)$ does not fork over C. By the Kim–Pillay theorem (see Fact 2.2), every relation with the properties listed there coincides with nonforking independence.

Fact 2.2 (Kim and Pillay). *Let $\mathop{\smile\!\!\!\!\!\!|}$ be a ternary relation on small subsets of $\mathbb{M}$, written $A \mathop{\smile\!\!\!\!\!\!|}_C B$, and assume that it satisfies the following:*

(i) invariance *under automorphisms of $\mathbb{M}$;*
(ii) symmetry*: the relation $A \mathop{\smile\!\!\!\!\!\!|}_C B$ holds if and only if $B \mathop{\smile\!\!\!\!\!\!|}_C A$ holds;*
(iii) monotonicity and transitivity*: the relation $A \mathop{\smile\!\!\!\!\!\!|}_C BD$ holds if and only if both $A \mathop{\smile\!\!\!\!\!\!|}_C B$ and $A \mathop{\smile\!\!\!\!\!\!|}_{CB} D$ hold;*
(iv) finite character*: the relation $A \mathop{\smile\!\!\!\!\!\!|}_C B$ holds if and only if $a \mathop{\smile\!\!\!\!\!\!|}_C b$ holds for all finite tuples a from A and b from B;*

(v) local character*: for every finite tuple a and every set B there is $B_0 \subseteq B$ with $|B_0| \leq |T|$ and $a \mathop{\smile\!\!\!\!\!\!|}_{B_0} B$, where $|T|$ is the number of formulas of the language of T;*
(vi) extension*: for every tuple a and all sets $C \subseteq B$ there is a' with $\mathrm{tp}(a'/C) = \mathrm{tp}(a/C)$ and $a' \mathop{\smile\!\!\!\!\!\!|}_C B$;*
(vii) independence theorem over models*: if M is a small elementary substructure of $\mathbb{M}$, $B \mathop{\smile\!\!\!\!\!\!|}_M D$, $a_1 \mathop{\smile\!\!\!\!\!\!|}_M B$, $a_2 \mathop{\smile\!\!\!\!\!\!|}_M D$ and $\mathrm{tp}(a_1/M) = \mathrm{tp}(a_2/M)$, then there is a with $\mathrm{tp}(a/MB) = \mathrm{tp}(a_1/MB)$, $\mathrm{tp}(a/MD) = \mathrm{tp}(a_2/MD)$ and $a \mathop{\smile\!\!\!\!\!\!|}_M BD$.*

Then the theory T is simple and the relation $\mathop{\smile\!\!\!\!\!\!|}$ coincides with nonforking independence (see [KP97, Theorem 4.2]*).*

3. The two paradigms

Here we prove Theorem 1.1(A): the Rado graph and the Pauli group on countably many qubits, taken modulo phases and with anticommutation as its relation, have simple unstable theories. In the subsections below we show that both are Fraïssé limits whose theories eliminate quantifiers, and that in both the independence relation of the Kim–Pillay theorem (see Fact 2.2) is read off from algebraic closure, which adds to a set the elements that lie in finite sets definable with parameters from it, and which takes each set to itself in the graph and to its linear span in the Pauli group.

3.1. **The Rado graph.** The class of finite graphs has the amalgamation property, since two graphs that share an induced subgraph can be glued along it without new edges. Its Fraïssé limit is the Rado graph (Γ, E), with vertex set Γ and edge relation E, and a countable graph is isomorphic to the Rado graph if and only if it has the *extension property*: for all finite disjoint sets Γ_1 and Γ_0 of vertices, there are infinitely many vertices outside $\Gamma_1 \cup \Gamma_0$ adjacent to every vertex of Γ_1 and to no vertex of Γ_0 [Fra54, Rad64]. The theory of (Γ, E) is $\aleph_0$-categorical and eliminates quantifiers, so the type of a tuple over a set D is determined by the equalities and edges among its entries and between its entries and the elements of D. By saturation, the monster model $\mathbb{M}$ of the theory of (Γ, E) has the extension property for small disjoint sets Γ_1 and Γ_0 of vertices.

The relation $E(x, y)$ has the independence property, since, by compactness, it suffices to realize every finite pattern: for every positive integer n, all distinct vertices $a_1, \ldots, a_n$ of the Rado graph and every set of indices among $1, \ldots, n$, the extension property gives a vertex adjacent to a_i exactly when i is in the set. Hence the theory of the Rado graph is unstable.

Theorem 3.1. *The theory of the Rado graph is simple, and, for small sets A, B and C of vertices of the monster model $\mathbb{M}$ of this theory, nonforking independence $A \mathop{\smile\!\!\!\!\!\!|}_C B$ holds if and only if $A \cap B \subseteq C$.*

Proof. We verify the hypotheses of Fact 2.2 for the relation $A \mathop{\smile\!\!\!\!\!\!|}_C B$ defined by $A \cap B \subseteq C$. The first five are set algebra, and extension and the independence theorem follow from the extension property of $\mathbb{M}$. Invariance, symmetry and finite character are immediate. Monotonicity and transitivity hold together, since the conditions $A \cap B \subseteq C$ and $A \cap D \subseteq C \cup B$ say jointly that every element of $A \cap (B \cup D)$ lies in C. For local character, given a finite tuple a and a set B, the finite set $B_0 = B \cap a$ satisfies $a \cap B \subseteq B_0$, which is the relation $a \mathop{\smile\!\!\!\!\!\!|}_{B_0} B$.

For extension, let a and $C \subseteq B$ be given, and write $a = (c, a_0)$, where c lists the entries of a in C. The type of a_0 over C prescribes only the equalities and edges of a_0 within itself and to C. Hence the extension property of $\mathbb{M}$, applied one entry at a time, with $B \setminus C$ added to

the set of vertices to which nonadjacency is prescribed, gives a tuple a'_0 disjoint from B with the same type over C, and the tuple $a' = (c, a'_0)$ satisfies $a' \cap B \subseteq C$.

For the independence theorem, let M, B, D, a_1 and a_2 be as in Fact 2.2(vii). Since $\mathrm{tp}(a_1/M) = \mathrm{tp}(a_2/M)$, the two tuples have the same entries in M and the same remaining equalities; write $a_\ell = (m, a'_\ell)$ for $\ell = 1, 2$, where the tuple m lists these common entries and each a'_ℓ is disjoint from M. The hypotheses $a_1 \mathop{\smile\!\!\!\!\!\!|}_M B$ and $a_2 \mathop{\smile\!\!\!\!\!\!|}_M D$ say that a'_1 is disjoint from B and a'_2 from D. We look for a tuple x, disjoint from $M \cup B \cup D$, whose equalities and edges within itself and edges to M are those of a'_1 (equivalently, of a'_2), whose edges to $B \setminus M$ are those of a'_1, and whose edges to $D \setminus M$ are those of a'_2. These prescriptions concern disjoint sets of vertices, because $B \cap D \subseteq M$; the extension property of $\mathbb{M}$ therefore provides such an x, and by quantifier elimination the tuple $a = (m, x)$ is as required. □

3.2. **Pauli strings.** The Hilbert space of a qubit is $\mathbb{C}^2$, and the bit flip and the phase flip are two of the Pauli matrices, which anticommute:

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad XZ = -ZX.$$

For a positive integer n, a Pauli string on n qubits is, up to a phase, a tensor product $X^{x_1}Z^{z_1} \otimes \cdots \otimes X^{x_n}Z^{z_n}$ with x_k and z_k in the two-element field $\mathbb{F}_2$ for $k = 1, \ldots, n$, and it is labeled by the vector $(x, z) \in \mathbb{F}_2^{2n}$, where $x = (x_1, \ldots, x_n)$ and $z = (z_1, \ldots, z_n)$; products of strings correspond to sums of labels. For a label $v = (x, z)$ we write P_v for the Hermitian string $\mathrm{i}^{N(v)}X^{x_1}Z^{z_1} \otimes \cdots \otimes X^{x_n}Z^{z_n}$, where $N(v)$ is the number of indices k with $x_k = z_k = 1$; it satisfies $P_v^2 = I$, where I is the identity, and the label $(1, 1)$ of one qubit gives the third Pauli matrix $P_{(1,1)} = \mathrm{i}XZ = Y$. Two strings either commute or anticommute, and

$$P_{(x,z)}P_{(x',z')} = (-1)^{\omega((x,z),(x',z'))}P_{(x',z')}P_{(x,z)}, \qquad \omega((x, z), (x', z')) = x \cdot z' + z \cdot x',$$

where $x \cdot z' = \sum_{k=1}^{n} x_k z'_k$, computed in $\mathbb{F}_2$, as in the Introduction. The form ω is bilinear, alternating and nondegenerate on $\mathbb{F}_2^{2n}$. Recall that in error correction an error is a Pauli string that acts on an encoded state as noise, and a check is a Pauli string whose measurement detects errors; as in the Introduction, we identify each Pauli string, up to its phase, with its label. The *syndrome* of an error e with respect to checks $g_1, \ldots, g_m$, for a positive integer m, is the vector $(\omega(g_1, e), \ldots, \omega(g_m, e))$: when the checks commute pairwise, measuring them on a state that they fix, after the error has acted on it, reveals which of them anticommute with the error. On countably many qubits we write V for the direct sum $\bigoplus_{k \in \mathbb{N}} \mathbb{F}_2^2$, with the form given by the same formula, now a finite sum for each pair of vectors; its elements are the labels of the Pauli strings that act nontrivially on only finitely many qubits.

3.3. **The Pauli group as a Fraïssé limit.** In the language with $+$, 0 and a binary relation for $\omega(x, y) = 1$, substructures are $\mathbb{F}_2$-subspaces with the restricted form. Finite alternating spaces over $\mathbb{F}_2$, the finite vector spaces over $\mathbb{F}_2$ with an alternating bilinear form, amalgamate: if $W_0 \subseteq W_1$ and $W_0 \subseteq W_2$, write $W_1 = W_0 \oplus W'_1$ and $W_2 = W_0 \oplus W'_2$, and give $W_0 \oplus W'_1 \oplus W'_2$ the bilinear form that extends the forms of W_1 and W_2 and vanishes on $W'_1 \times W'_2$ and on $W'_2 \times W'_1$. Every finite alternating space embeds in a nondegenerate one, and by Witt's theorem [Art57] every isometry between subspaces of a finite nondegenerate alternating space extends to an isometry of the whole space. A back-and-forth argument along symplectic bases therefore shows that (V, ω) is the Fraïssé limit of the finite alternating spaces over $\mathbb{F}_2$. Its theory is $\aleph_0$-categorical and eliminates quantifiers, so the type of a tuple a over a set D is determined by the linear relations of a over $\mathrm{span}(D)$ and the values of ω on $\mathrm{span}(aD)$. The algebraic closure of a set is its span.

Lemma 3.2. *Let n be a positive integer. If $a_1, \dots, a_n \in V$ are linearly independent and $\varepsilon_1, \dots, \varepsilon_n \in \mathbb{F}_2$, then there is $b \in V$ with $\omega(a_i, b) = \varepsilon_i$ for $i = 1, \dots, n$.*

Proof. The vectors a_i lie in the span W of the labels of X and Z on the first m qubits, for a large enough integer m, and the form is nondegenerate on W. Hence the functionals $\omega(a_i, \cdot)$ restricted to W are linearly independent, and the linear map $b \mapsto (\omega(a_i, b))_{i \le n}$ from W to $\mathbb{F}_2^n$ is onto. □

By Lemma 3.2 and compactness, the formula $\omega(x, y) = 1$ has the independence property: every syndrome with respect to finitely many linearly independent checks is the syndrome of some error. Hence the theory of (V, ω) is unstable. By saturation, for every small subspace D of the monster model $\mathbb{M}$ of the theory of (V, ω) and every positive integer n, every alternating space $D \oplus \operatorname{span}(x_1, \dots, x_n)$ that extends D by n linearly independent vectors is realized over D by elements of $\mathbb{M}$; in particular, for every linear functional f on D there is $b \in \mathbb{M}$ with $\omega(\cdot, b) = f$ on D.

Theorem 3.3. *The theory of (V, ω) is simple, and, for small subsets A, B and C of the monster model $\mathbb{M}$ of this theory, nonforking independence $A \mathop{\smile\!\!\!\!\!\!|}_C B$ holds if and only if $\operatorname{span}(AC) \cap \operatorname{span}(BC) = \operatorname{span}(C)$.*

Proof. The proof follows that of Theorem 3.1, with linear spans in place of the sets themselves: the first five axioms are linear algebra, and extension and the independence theorem come from the freedom to prescribe the form on a complement. We verify the hypotheses of Fact 2.2 for the relation $A \mathop{\smile\!\!\!\!\!\!|}_C B$ defined by $\operatorname{span}(AC) \cap \operatorname{span}(BC) = \operatorname{span}(C)$. Invariance and symmetry are immediate, and finite character holds because a vector of $\operatorname{span}(AC) \cap \operatorname{span}(BC)$ outside $\operatorname{span}(C)$ is a combination of finitely many elements of A, B and C.

For monotonicity and transitivity, put $Q_1 = \operatorname{span}(AC)$, $Q_2 = \operatorname{span}(BC)$, $Q_3 = \operatorname{span}(BCD)$ and $Q_0 = \operatorname{span}(C)$; these subspaces satisfy $Q_2 \subseteq Q_3$ and $\operatorname{span}(ABC) = Q_1 + Q_2$. It remains to be shown that $Q_1 \cap Q_3 = Q_0$ if and only if $Q_1 \cap Q_2 = Q_0$ and $(Q_1 + Q_2) \cap Q_3 = Q_2$. Suppose that $Q_1 \cap Q_3 = Q_0$. Since $Q_2 \subseteq Q_3$, the intersection $Q_1 \cap Q_2$ is contained in Q_0. If $u_1 + u_2 \in Q_3$ with $u_1 \in Q_1$ and $u_2 \in Q_2$, then the vector u_1 also lies in Q_3, hence in $Q_1 \cap Q_3 = Q_0 \subseteq Q_2$; thus the sum lies in Q_2. Conversely, every $u \in Q_1 \cap Q_3$ lies in $(Q_1 + Q_2) \cap Q_3 = Q_2$, and therefore in $Q_1 \cap Q_2 = Q_0$.

Local character follows from finite dimension. Given a finite tuple a and a set B, the space $\operatorname{span}(a) \cap \operatorname{span}(B)$ is finite, so it is contained in $\operatorname{span}(B_0)$ for some finite $B_0 \subseteq B$. If $u + w \in \operatorname{span}(B)$ with $u \in \operatorname{span}(a)$ and $w \in \operatorname{span}(B_0)$, then the vector u lies in $\operatorname{span}(a) \cap \operatorname{span}(B) \subseteq \operatorname{span}(B_0)$. It follows that $\operatorname{span}(aB_0) \cap \operatorname{span}(B) = \operatorname{span}(B_0)$, which is the relation $a \mathop{\smile\!\!\!\!\!\!|}_{B_0} B$.

For extension, let a and $C \subseteq B$ be given. Replacing a by a maximal subtuple linearly independent over $\operatorname{span}(C)$, we may assume that a is linearly independent over $\operatorname{span}(C)$, since the remaining entries are then determined. Choose a complement B' of $\operatorname{span}(C)$ in $\operatorname{span}(BC)$. By saturation, for a tuple x of new vectors of the same length as a, the alternating space $\operatorname{span}(BC) \oplus \operatorname{span}(x)$ in which x has the values of ω prescribed by $\operatorname{tp}(a/C)$ on $\operatorname{span}(xC)$ and is orthogonal to B' is realized over $\operatorname{span}(BC)$ by a tuple a'; this tuple has the type of a over C and is linearly independent over $\operatorname{span}(BC)$, whence $\operatorname{span}(a'C) \cap \operatorname{span}(BC) = \operatorname{span}(C)$, which is the relation $a' \mathop{\smile\!\!\!\!\!\!|}_C B$.

The independence theorem reduces to a gluing of functionals. Let M, B, D, a_1 and a_2 be as in Fact 2.2(vii). As in the proof of extension, we may assume that a_1 is linearly independent over M; then so is a_2, and the hypotheses $a_1 \mathop{\smile\!\!\!\!\!\!|}_M B$ and $a_2 \mathop{\smile\!\!\!\!\!\!|}_M D$ make a_1 linearly independent over $\operatorname{span}(MB)$ and a_2 over $\operatorname{span}(MD)$. Let n be the common length

of a_1 and a_2, and, for $\ell = 1, 2$ and $k = 1, \ldots, n$, let $a_{\ell,k}$ be the kth entry of a_ℓ. For each k, the functionals $\omega(a_{1,k}, \cdot)$ on $\mathrm{span}(MB)$ and $\omega(a_{2,k}, \cdot)$ on $\mathrm{span}(MD)$ agree on M, because $\mathrm{tp}(a_1/M) = \mathrm{tp}(a_2/M)$. Since $B \mathop{\smile\!\!\!\!\!\!|}_M D$, their domains meet in $\mathrm{span}(MB) \cap \mathrm{span}(MD) = M$, and the two functionals glue to a linear functional on $\mathrm{span}(MBD)$. Realize over $\mathrm{span}(MBD)$ a tuple a, linearly independent over $\mathrm{span}(MBD)$, with these functionals and with the values of ω on a given by $\mathrm{tp}(a_1/M)$. By quantifier elimination, the type of a over MB is that of a_1, and its type over MD is that of a_2; the relation $a \mathop{\smile\!\!\!\!\!\!|}_M BD$ holds by construction. □

In both structures, then, nonforking independence is the independence given by algebraic closure: for small sets A, B and C, the relation $A \mathop{\smile\!\!\!\!\!\!|}_C B$ holds exactly when the closures of AC and BC meet in the closure of C. The independence theorem then reduces to the freedom to prescribe the remaining relations, edges or values of the form, on each side separately.

Remark 3.4. The proof of Theorem 3.3 applies over every finite field $\mathbb{F}_q$, for a prime power q, in the language with a unary function for multiplication by each scalar and a binary relation for each value of the form. For symmetric forms in characteristic two, and for arbitrary bilinear forms, the identification of the limit through symplectic bases and Witt's theorem is no longer available, but the Fraïssé limit still exists, since the finite spaces with such a form have free amalgamation. The verification of Fact 2.2 goes through with the values $\omega(x, x)$ prescribed by the type and, for forms that are not symmetric, with the functionals $\omega(x, \cdot)$ and $\omega(\cdot, x)$ glued separately.

4. Stabilizer codes as a structure: a dictionary

Three notions of model theory, namely Fraïssé limits, forking and the independence theorem, are already at work in quantum error correction under other names, and in this section we identify them. After a brief account of stabilizer codes, in each of the three subsections that follow it we state a theorem, prove it, and interpret it twice: once for the physicist, who knows stabilizer codes but not model theory, and once for the logician, who knows model theory but not stabilizer codes. A single example runs through the section: the Laflamme–Miquel–Paz–Zurek five-qubit code [LMPZ96], the smallest code that corrects an arbitrary error on one qubit, which Bennett, DiVincenzo, Smolin and Wootters [BDSW96] found independently. The dictionary is summarized in Table 1.

4.1. **Stabilizer codes in brief.** We recall what the logician needs (see [Got97]). For a positive integer n, a *stabilizer group* on n qubits is a group $\mathcal{G}$ of pairwise commuting Pauli strings that does not contain $-I$. Its common eigenspace for the eigenvalue 1 is the *code space*; if $\mathcal{G}$ has $n - k$ independent generators, for an integer k with $0 \le k \le n$, then the code space has dimension 2^k and encodes k qubits. In the labels of Section 3, the group $\mathcal{G}$ corresponds, up to signs, to an isotropic subspace $S \subseteq \mathbb{F}_2^{2n}$, that is, a subspace on which the symplectic form ω vanishes. An error is a Pauli string e; measuring independent generators $g_1, \ldots, g_{n-k}$ of $\mathcal{G}$ on the corrupted state returns the *syndrome* $(\omega(g_1, e), \ldots, \omega(g_{n-k}, e))$, and the decoder guesses the error from the syndrome. An error with zero syndrome is undetectable. It is harmless if it lies in $\mathcal{G}$ up to a phase, since it then acts on the code space as a scalar, and it is a *logical error* otherwise. On labels, the errors with zero syndrome form the *centralizer* $S^\perp = \{e \in \mathbb{F}_2^{2n} : \omega(s, e) = 0 \text{ for all } s \in S\}$, which contains S and has dimension $2n - (n - k) = n + k$; a set L of *logical operators* is a set of $2k$ labels with $S^\perp = S \oplus \mathrm{span}(L)$, and the values $\omega(\ell, e)$ for $\ell \in L$ record which logical operator an error of $S^\perp$ applies to the encoded qubits. If $k \ge 1$, the *distance* of the code is the least number of qubits on which a logical error acts. A *Clifford unitary* is a unitary operator U that maps Pauli strings to Pauli strings under

Model theory	**Quantum error correction**	
the structure (V, ω)	Pauli strings on countably many qubits, with anticommutation	Section 3
the age of (V, ω), its finite substructures up to isomorphism	the commutation matrices of finite sets of Pauli strings with linearly independent labels	Proposition 4.2
ultrahomogeneity, automorphisms	Clifford circuits; encoding circuits of stabilizer codes	Theorem 4.3
algebraic closure of a set	the group generated by a set of Pauli strings	Section 3
the type of an error over a set	its syndrome, and how, if at all, it is a product of elements of the set	Theorem 4.5
a forking extension	the error becomes a product of known operators	Theorem 4.5
nonforking over $\emptyset$ of the type over the stabilizer group, for a nonzero label with zero syndrome	an undetectable error that is not harmless	Corollary 4.6
forking over the stabilizer group of the type over the stabilizer group and a set of logical operators	a logical error	Proposition 4.7
the independence theorem	syndromes of independent checks can be prescribed independently	Proposition 4.8
idealized pure probes	Pauli strings of arbitrary support	Proposition 5.4
probes	Pauli channels	Proposition 5.4
the difference of the responses of the uniform mixtures on the two cosets of a subgroup of index 2	the projector onto the states of a given syndrome	Proposition 7.1

TABLE 1. The Pauli group as a structure: the dictionary of Sections 4, 5 and 7.

conjugation, up to a phase; on labels, conjugation by U is a linear map of $\mathbb{F}_2^{2n}$ that preserves ω.

Example 4.1. The five-qubit code has the stabilizer generators

$$g_1 = XZZXI, \quad g_2 = IXZZX, \quad g_3 = XIXZZ, \quad g_4 = ZXIXZ,$$

where each word stands for the tensor product of its letters (thus $XZZXI$ stands for $X \otimes Z \otimes Z \otimes X \otimes I$), the letters being the Pauli matrices X, Z and $Y = \mathrm{i}XZ$ of Section 3 and the identity I of $\mathbb{C}^2$. The code encodes one qubit, with logical operators $\bar{X} = XXXXX$ and $\bar{Z} = ZZZZZ$. Each of the fifteen errors that act as X, Y or Z on a single qubit has its own nonzero syndrome, and there are exactly fifteen nonzero syndromes of four bits; for instance, the error XIIII has syndrome $(0, 0, 0, 1)$ and the error $IIZII$ has syndrome $(0, 0, 1, 0)$. The code therefore corrects every error on one qubit, and its distance is at least

3; after Proposition 4.7 we exhibit a logical error that acts on exactly three qubits, which shows that the distance is exactly 3.

4.2. **Fraïssé limits and Clifford circuits.** In Proposition 4.2 and Theorem 4.3 below, we show that every finite piece of commutation structure occurs among Pauli strings, and that every symmetry between finite pieces is implemented by a Clifford circuit, a product of elementary Clifford unitaries on one or two qubits that serve as the gates of a quantum computer; every Clifford unitary is such a product, up to a phase (see [Got97, DDM03]).

Proposition 4.2. *Let d be a positive integer. If Θ is a symmetric $d \times d$ matrix over $\mathbb{F}_2$ with zero diagonal, then there are Pauli strings $P_1, \dots, P_d$ with linearly independent labels, on at most d qubits, such that, for all $i, j \in \{1, \dots, d\}$, the strings P_i and P_j anticommute exactly when $\Theta_{ij} = 1$.*

Proof. The matrix Θ defines an alternating form on $\mathbb{F}_2^d$. Let W_0 be its radical and choose a complement W of W_0; the form is nondegenerate on W, which therefore has a symplectic basis of r pairs, where $2r = \dim W$. Send the r pairs to the labels of X and Z on the first r qubits, and a basis of W_0 to the labels of X on $\dim W_0$ further qubits. This linear map is injective and preserves the form, and the images of the standard basis vectors of $\mathbb{F}_2^d$ are the required strings, on $r + \dim W_0 \leq d$ qubits. □

Theorem 4.3. *Let n be a positive integer, let W and W' be subspaces of $\mathbb{F}_2^{2n}$, and let $f \colon W \to W'$ be a linear bijection with $\omega(f(v), f(w)) = \omega(v, w)$ for all $v, w \in W$. Then there is a Clifford unitary U on n qubits with $UP_vU^* = \pm P_{f(v)}$ for all $v \in W$, where P_v is the Hermitian Pauli string with label v (see Section 3). If, moreover, the subspace W is isotropic and the vectors $v_1, \dots, v_m$ form a basis of W, then the unitary U can be chosen with $UP_{v_i}U^* = P_{f(v_i)}$ for all $i \leq m$.*

Proof. By Witt's theorem, the map f extends to a linear bijection τ of $\mathbb{F}_2^{2n}$ that preserves ω. Every such τ is induced by conjugation by a Clifford unitary [CRSS98, DDM03]; let U be one. Since the operator UP_vU^* is a Pauli string up to a phase, Hermitian and with square I, the phase is a sign. For the second statement, let $\varepsilon_i \in \{\pm 1\}$ be the sign with $UP_{v_i}U^* = \varepsilon_i P_{f(v_i)}$. The vectors $f(v_i)$ are linearly independent, so by nondegeneracy there is $b \in \mathbb{F}_2^{2n}$ with $(-1)^{\omega(f(v_i), b)} = \varepsilon_i$ for all i. Conjugation by P_b multiplies $P_{f(v_i)}$ by $(-1)^{\omega(f(v_i), b)}$, so the Clifford unitary P_bU has the required property. □

Corollary 4.4. *Let n be a positive integer. Any two stabilizer groups on n qubits with the same number of independent generators are conjugate by a Clifford unitary. In particular, for every integer k with $0 \leq k \leq n$, every stabilizer code that encodes k qubits into n has an encoding circuit made of Clifford gates: a Clifford unitary that maps the code space of the trivial code, with generators $Z_1, \dots, Z_{n-k}$, where Z_i is the phase flip on the ith qubit, onto the code space of the given code.*

Proof. Let $\mathcal{G}$ and $\mathcal{G}'$ be the two stabilizer groups, let m be their common number of independent generators, and let $g_1, \dots, g_m$ and $g'_1, \dots, g'_m$ be independent generators of $\mathcal{G}$ and of $\mathcal{G}'$, respectively, with labels v_i and v'_i for $i = 1, \dots, m$. Both families span isotropic subspaces, so the linear map $v_i \mapsto v'_i$ preserves ω, and Theorem 4.3 gives a Clifford unitary U_0 with $U_0 P_{v_i} U_0^* = P_{v'_i}$ for all i. The square of each generator is a scalar, I or $-I$, that lies in its group, and it is therefore I, since a stabilizer group does not contain $-I$; hence $g_i = \varepsilon_i P_{v_i}$ and $g'_i = \varepsilon'_i P_{v'_i}$ for signs $\varepsilon_i, \varepsilon'_i \in \{\pm 1\}$. As in the proof of Theorem 4.3, there is $b \in \mathbb{F}_2^{2n}$ with

$(-1)^{\omega(v_i',b)} = \varepsilon_i\varepsilon_i'$ for all i, and the Clifford unitary $U = P_bU_0$ satisfies $Ug_iU^* = g_i'$ for all i, whence $U\mathcal{G}U^* = \mathcal{G}'$. □

On countably many qubits, every finite-dimensional subspace of V lies in the labels of the first n qubits for some n. Applying Theorem 4.3 there, and extending by the identity on the remaining qubits, shows that every isometry between finite-dimensional subspaces of (V,ω) extends to an automorphism of (V,ω) implemented by a Clifford circuit on finitely many qubits.

For the physicist or operator theorist. A Fraïssé limit is a countable structure whose finite configurations are, up to isomorphism, exactly those of a given kind and in which every symmetry between two finite configurations extends to a symmetry of the whole structure; any two such structures are isomorphic. By Proposition 4.2, every finite pattern of commutation relations occurs among Pauli strings, and by Theorem 4.3, every symmetry between finite pieces extends to a global one, implemented by a Clifford circuit. The Pauli group on countably many qubits is therefore the Fraïssé limit of the finite alternating spaces over $\mathbb{F}_2$, that is, of finite commutation patterns together with the linear relations among the labels, and Clifford circuits on finitely many qubits implement the symmetries that witness its homogeneity. The existence of encoding circuits for stabilizer codes, usually proved by an explicit construction, is the shadow of this homogeneity: by Corollary 4.4, the five-qubit code is the image of the trivial code with generators Z_1, Z_2, Z_3, Z_4, the phase flips on the first four qubits, under a Clifford circuit, because both groups have four independent generators, all commuting, and the two generating sets therefore have the same commutation pattern.

For the logician. The structure (V,ω) of Section 3 is the Pauli group modulo phases, and, for each positive integer n, the Clifford unitaries on the first n qubits act on the finite piece $(\mathbb{F}_2^{2n},\omega)$ through its automorphism group, the symplectic group. Ultrahomogeneity, which in Section 3 came from Witt's theorem and a back-and-forth argument, is implemented by Clifford circuits. On labels, a stabilizer code becomes an isotropic subspace, which determines it up to the signs of its generators, and Corollary 4.4 follows from ultrahomogeneity for isotropic subspaces, with the signs of the generators as additional data handled by Lemma 3.2.

4.3. **Forking and logical errors.** In a simple theory, an extension of the type of an element forks when, in the sense recalled in Section 2 and explained after Proposition 4.7, it confines the element to one of infinitely many alternatives that the original type cannot tell apart. The next theorem computes forking for the Pauli group, using the description of nonforking independence in Theorem 3.3.

Theorem 4.5. *Let e be an element of a monster model $\mathbb{M}$ of the theory of (V,ω), and let $D \subseteq B$ be small subsets of $\mathbb{M}$. Then the type $\mathrm{tp}(e/B)$ forks over D if and only if $e \in \mathrm{span}(B) \setminus \mathrm{span}(D)$.*

Proof. By Theorem 3.3, the type $\mathrm{tp}(e/B)$ does not fork over D exactly when $\mathrm{span}(eD) \cap \mathrm{span}(B) = \mathrm{span}(D)$. If $e \in \mathrm{span}(D)$, the spans $\mathrm{span}(eD)$ and $\mathrm{span}(D)$ coincide, and the equality holds. Otherwise the span of eD is $\mathrm{span}(D) \oplus \{0, e\}$, and the equality fails if and only if some vector $e + d$ with $d \in \mathrm{span}(D)$ lies in $\mathrm{span}(B)$, that is, if and only if $e \in \mathrm{span}(B)$. □

By Theorem 4.5, commutation data never fork: the type of an element e of $\mathbb{M}$ over a small set B records the values $\omega(b,e)$ for $b \in B$, and these values, constrained only by linearity and by the type of e over a subset D of B, can be prescribed without forking over D. Forking happens only when e becomes a linear combination of elements of B; for Pauli strings, this

means that e becomes, up to a phase, a product of strings in B. In a stabilizer code this criterion separates the two kinds of nonzero undetectable error. To state the consequence for the distance of a code, we call the number of qubits on which an error acts nontrivially its *weight*.

Corollary 4.6. *Let n be a positive integer, and regard $\mathbb{F}_2^{2n}$ as the subspace of V formed by the labels of the strings on the first n qubits. Let $\mathcal{G}$ be a stabilizer group on n qubits with set of labels S, and let $e \in \mathbb{F}_2^{2n}$ be a nonzero label with zero syndrome. Then the type $\mathrm{tp}(e/S)$ forks over $\emptyset$ if and only if e is harmless, that is, lies in S. Consequently, if the code of $\mathcal{G}$ encodes at least one qubit, its distance is the least weight of a nonzero label in $\mathbb{F}_2^{2n}$ with zero syndrome whose type over S does not fork over $\emptyset$.*

Proof. Apply Theorem 4.5 with $D = \emptyset$ and $B = S$; the span of S is S itself, and $\mathrm{span}(\emptyset) = \{0\}$. The second statement is the definition of distance, since a logical error is a nonzero label with zero syndrome outside S. □

In Proposition 4.7 below, we show that the stabilizer group alone does not tell logical errors apart, because they all have the same type over its set of labels, and that the logical operators, once known, characterize them by forking.

Proposition 4.7. *Let n be a positive integer, regard $\mathbb{F}_2^{2n}$ as a subspace of V as in Corollary 4.6, let $\mathcal{G}$ be a stabilizer group on n qubits with set of labels S, let L be a set of logical operators of its code, and let $e \in \mathbb{F}_2^{2n}$ be a label with zero syndrome. Then the following hold:*

(a) *the label e is a logical error if and only if $\mathrm{tp}(e/S \cup L)$ forks over S;*
(b) *all logical errors have the same type over S, and this type does not fork over $\emptyset$.*

Proof. (a) Since the syndrome of e is zero, the label e lies in $S^\perp = \mathrm{span}(S \cup L)$. By Theorem 4.5 with $D = S$ and $B = S \cup L$, the type $\mathrm{tp}(e/S \cup L)$ forks over S if and only if $e \in \mathrm{span}(S \cup L) \setminus S = S^\perp \setminus S$, which is the set of logical errors.

(b) Suppose that the label e is a logical error, that is, $e \in S^\perp \setminus S$. By quantifier elimination, the type of e over S is determined by the linear relations of e over S, of which there are none since $e \notin S$, and by the values of ω on $\mathrm{span}(eS)$. These values are fixed, through bilinearity, by the values of ω on S, by $\omega(e, e)$, which is 0 because ω is alternating, and by the values $\omega(s, e)$ for $s \in S$, which are 0 because e lies in $S^\perp$. Hence all logical errors have the same type over S. The label e is nonzero, since the zero label lies in S; as it has zero syndrome and lies outside S, its type over S does not fork over $\emptyset$, by Corollary 4.6. □

In the five-qubit code, the errors $g_1 g_2 = XYIYX$ (up to a phase) and $\bar{X} = XXXXX$ both have zero syndrome. The first lies in the stabilizer group, whose set of labels we denote by S, so its type over S forks over $\emptyset$; it is harmless. The second does not, so its type does not fork over $\emptyset$; it flips the encoded qubit, and it acts on five qubits. By Example 4.1, an error of the second kind acts on at least three qubits, and the logical error $\bar{X} g_1 = IYYIX$, up to a phase, acts on exactly three; the distance of the code is therefore 3. With $L = \{\bar{X}, \bar{Z}\}$, the logical errors $\bar{X}$, $\bar{Z}$ and $\bar{X} g_1$ have the same type over S; over $S \cup L$ each of them lies in $\mathrm{span}(S \cup L)$, whence its type becomes algebraic and forks over S, as Proposition 4.7(a) predicts.

For the physicist or operator theorist. Forking is the model-theoretic notion of an observation that singles out one of infinitely many alternatives that the data known in advance cannot tell apart and of which, for some fixed integer $k \geq 2$, no k can hold together. To be precise, in a simple theory, by a theorem of Kim [Kim98], a condition $\varphi(x, b)$ on an unknown

element x forks over a set D of known parameters exactly when, for some $k \geq 2$, the parameter b is the first term of a sequence $(b_j)_{j\in\mathbb{N}}$ that is indiscernible over D and k-inconsistent for φ, in the sense that no k of the conditions $\varphi(x, b_j)$ can hold together (see Definition 2.1). A sequence is indiscernible over D if any two of its increasing finite subsequences of the same length satisfy the same formulas with parameters from D. A type forks over D when one of its conditions does; such a type narrows the possibilities for the element but need not determine it. In the Pauli group a forking type does determine the element. The type of an error over a set B of known operators records the syndrome of the error with respect to B and whether, and how, the error is a product of operators in B; by Theorem 4.5, this type forks over a subset D of B exactly when the error is a product of operators in B but not of operators in D. Syndromes, by contrast, only narrow: on infinitely many qubits, no finite set of checks determines an error, however many bits it reveals. In a code, with S the set of labels of its stabilizer group, this criterion separates the two kinds of undetectable error other than the identity (see Corollary 4.6): the type over S of a harmless one, which is a product of stabilizers, forks over the empty set, whereas that of a logical error does not. The distance of the code, its single most important parameter, is therefore the least weight of an undetectable error on the qubits of the code, other than the identity, whose type over S does not fork over the empty set. The picture is completed by Proposition 4.7: the stabilizer group alone sees all logical errors alike, and each logical error is pinned down as soon as a set L of logical operators joins the known operators, since its type over $S \cup L$ forks over S.

For the logician. The theory of (V, ω) has a vector space pregeometry for algebraic closure, and by Theorem 4.5 its forking is the dependence of this pregeometry: the type of a single element over a set forks over a subset exactly when the element is algebraic over the set but not over the subset. Thus the theory has SU-rank one, as does the theory of the Rado graph (see Theorem 3.1), with linear span in place of equality. In finite dimension, the information carried by a syndrome is measured by counting: if n and m are positive integers and $s_1, \ldots, s_m$ are linearly independent labels in $\mathbb{F}_2^{2n}$, then for each syndrome $\sigma \in \mathbb{F}_2^m$ exactly 2^{2n-m} labels e satisfy $\omega(s_i, e) = \sigma_i$ for all i, since the map $e \mapsto (\omega(s_i, e))_{i\leq m}$ is linear and onto. This count records codimension, which forking ignores. With S the set of labels of the stabilizer group of a code, the logical errors of the code are the elements of the centralizer of S outside S, and by Proposition 4.7 they share one type over S. For a set L of logical operators, the nonforking extensions of this type to $S \cup L$ are realized in (V, ω) by labels outside $\operatorname{span}(S \cup L)$ that commute with S, for instance by errors on qubits that the code does not touch; the logical errors realize the forking extensions of the same type, which become algebraic once the logical operators are added.

4.4. **The independence theorem and the gluing of syndromes.** The Kim–Pillay independence theorem says that two nonforking extensions of a type over a model, to sets that are independent over the model, have a common nonforking extension. In the Pauli group it becomes a statement about syndromes.

Proposition 4.8. *Let n be a positive integer, let C, B and D be subsets of $\mathbb{F}_2^{2n}$ with $\operatorname{span}(BC) \cap \operatorname{span}(DC) = \operatorname{span}(C)$, and let ξ_1 and ξ_2 be linear functionals on $\operatorname{span}(BC)$ and on $\operatorname{span}(DC)$, respectively, that agree on $\operatorname{span}(C)$. Then there is a label $e \in \mathbb{F}_2^{2n}$ with $\omega(s, e) = \xi_1(s)$ for $s \in \operatorname{span}(BC)$ and $\omega(s, e) = \xi_2(s)$ for $s \in \operatorname{span}(DC)$.*

Proof. Since ξ_1 and ξ_2 agree on the intersection of their domains, there is a linear functional ξ on $\operatorname{span}(BC) + \operatorname{span}(DC)$ extending both. Extend ξ to $\mathbb{F}_2^{2n}$. Because ω is nondegenerate, every linear functional on $\mathbb{F}_2^{2n}$ has the form $\omega(\cdot, e)$ for a unique e. □

In the five-qubit code, take $C = \emptyset$, $B = \{g_1, g_2\}$ and $D = \{g_3, g_4\}$. The spans of B and D meet only in 0, so any two bits of syndrome on g_1, g_2 and any two on g_3, g_4 are realized together by one error; by Example 4.1, every nonzero syndrome is realized even by an error on a single qubit, for instance the syndrome $(1, 0 \,|\, 1, 1)$ by $YIIII$. If D is replaced by $\{g_1 g_2\}$, the hypothesis fails, and so does the conclusion: the syndrome bit of every error with respect to $g_1 g_2$ is the sum of its bits with respect to g_1 and g_2.

For the physicist or operator theorist. The independence theorem is the rule that lets partial information be combined: two sets of checks that are independent over a common set constrain each other only on that set: two syndromes prescribed separately on them are realized by a single error as soon as they agree on the common set. The point of the Kim–Pillay theorem is that this gluing rule, together with the other properties of Fact 2.2, characterizes the simple theories [KP97]. In Section 6 we show that the rule has no counterpart once preparations are mixed, for instance in the Pauli observable of Section 5, in which Alice holds a check and Bob's probe is a random error: for two halves of a finite set of linearly independent checks that meet in a quarter of the set, the condition that the error anticommutes almost surely with every check of one half and commutes almost surely with every other check of the set and the same condition for the other half are each consistent but cannot hold together; infinitely many such conditions produce the tree property of the second kind.

For the logician. The statement of Proposition 4.8 is the counterpart, in finite dimension and over the span of an arbitrary set rather than over a model, of the independence theorem verified in the proof of Theorem 3.3: the whole argument is the gluing of two functionals whose domains, the spans of two sets of checks together with a common set, meet only in the span of the common set. Its physical content is that the syndrome bits of an error with respect to a set of checks are the values of a linear functional on the span of the checks, and the consistency of prescribed syndromes is therefore a matter of linear algebra on that span; the surjectivity of the syndrome map of linearly independent checks is Lemma 3.2.

5. Observables, mixtures and dividing

We now transport the tree properties of Definition 2.1 to the responses of an observable, as we transported the independence property in the second paper of this series [DI26b], and introduce the observables of the two paradigms. Their probes, we find, are of different kinds (see Proposition 5.4 and Lemma 5.5): the probes of the Pauli observable are the Pauli channels, and its pure probes are the Pauli strings of arbitrary support; the pure probes of the Rado observable are all sets of neighbors.

5.1. **Diagonal observables.** Let A and B be countable sets and let $\varphi\colon A \times B \to \{0, 1\}$ be a function. The *diagonal observable* of φ is the operator

$$C_\varphi = \sum_{a \in A,\, b \in B} \varphi(a, b)\, |a\rangle\langle a| \otimes |b\rangle\langle b| \quad \text{on } \ell^2(A) \otimes \ell^2(B).$$

Here $|a\rangle\langle a|$ and $|b\rangle\langle b|$ are the orthogonal projections onto the lines spanned by the basis vectors of $\ell^2(A)$ and $\ell^2(B)$ indexed by a and b, respectively. In plain words, Alice's register holds a label a, Bob's a label b, and the observable reads $\varphi(a, b)$. A density operator ρ of Alice, that is, a positive operator of trace one on $\ell^2(A)$ that describes her preparation, enters only through its diagonal, which is a probability measure μ on A, a *mixture* of her labels: the preparation draws a at random with law μ. The operator on Bob's space obtained by

averaging C_φ over Alice's preparation, the effective observable of ρ in the terminology of the second paper, is the diagonal operator of multiplication by

$$b \longmapsto \sum_{a\in A} \mu(a)\varphi(a,b).$$

The probe algebra of the second paper is the unital C*-algebra generated by these effective observables, that is, the smallest algebra of operators on Bob's space that contains them and the identity and is closed under adjoints and under limits in the operator norm; it is the commutative C*-algebra of bounded functions on B generated by the constants and the indicators $b \mapsto \varphi(a,b)$, $a \in A$. Bob's probes are the states of the probe algebra, the positive linear functionals on it with value 1 at the identity, and they form the probe space, which carries the weak* topology, the topology of pointwise convergence on the probe algebra. The probe space is described through the closure K_φ of the set of functions $\varphi(\cdot,b)$, $b \in B$, in $\{0,1\}^A$ with the product topology.

Lemma 5.1. *Let A and B be countable sets and let $\varphi\colon A\times B \to \{0,1\}$ be a function. Then the probe algebra of the diagonal observable C_φ is isomorphic to the algebra $C(K_\varphi)$ of continuous complex functions on the closure K_φ of the set of functions $\varphi(\cdot,b)$, $b \in B$, in $\{0,1\}^A$, the probe space is the space $\mathcal{P}(K_\varphi)$ of Radon probability measures on K_φ with the weak* topology, and the response of a mixture μ, a probability measure on A, is the affine continuous function*

$$r_\mu(\nu) = \int_{K_\varphi} \sum_{a\in A} \mu(a)\lambda(a)\, d\nu(\lambda) \qquad (\nu \in \mathcal{P}(K_\varphi)).$$

Proof. The unital C*-algebra generated by commuting projections is the algebra of continuous functions on the Stone space of the Boolean algebra they generate. The Boolean algebra generated by the sets $\{b : \varphi(a,b) = 1\}$ has as Stone space the closure of the image of B under $b \mapsto \varphi(\cdot,b)$, which is K_φ. States of $C(K_\varphi)$ are Radon probability measures by the Riesz representation theorem, and the formula for r_μ is the evaluation of the state ν at the effective observable of a preparation with diagonal μ. □

The pure probes are the point masses δ_λ, $\lambda \in K_\varphi$, and we identify each of them with its point λ; a pure probe is thus an *idealized label* of Bob, recording which of Alice's labels it answers with 1. The actual labels, the functions $\varphi(\cdot,b)$ for $b \in B$, are dense among them, since K_φ is the closure of the set of these functions. For a label b, we write $r_\mu(b)$ for the response at the pure probe $\varphi(\cdot,b)$.

5.2. **The structure of an observable.** To speak of the theory of an observable, we regard it as a structure of real-valued logic. For a diagonal observable C_φ, let $\mathcal{S}_\varphi$ be the two-sorted structure whose sort of preparations consists of Alice's mixtures, the probability measures on A, whose sort of probes consists of Bob's probes, the space $\mathcal{P}(K_\varphi)$, and whose only predicate is the response $R(\mu,\nu) = r_\mu(\nu)$, with values in $[0,1]$. As in the second paper, we work in Keisler's real-valued logic [Kei23], in which no metric is given a priori; in an ultrapower, two elements of the same sort are identified when every atomic formula takes the same value at them, whatever the elements in its other places. We say that C_φ is *simple* if the real-valued theory of $\mathcal{S}_\varphi$ is simple in the sense of Ben Yaacov [BY03]. In an elementary extension of $\mathcal{S}_\varphi$, a larger structure in which every formula with parameters from $\mathcal{S}_\varphi$ takes the same value as in $\mathcal{S}_\varphi$ itself, the new elements of the sort of preparations are idealized preparations of Alice. A tree property of a formula, when the theory has it, is witnessed in every $\aleph_1$-saturated elementary extension, one in which every countable set of conditions with parameters is realized as soon

as each of its finite subsets is. When a tree property is witnessed in $\mathcal{S}_\varphi$ itself, in the sense of Definition 5.2 below, we say that actual preparations witness it.

5.3. **Dividing for families of functions.** The responses of an observable form a family of real functions on the probe space, and in the second paper we defined the independence property for such a family: a sequence in the family is independent when, for some reals $\alpha < \beta$, any two disjoint finite sets of indices admit a point at which the functions indexed by the first set take values at most α and those indexed by the second values at least β; the family has the independence property when it contains an independent sequence, and it is NIP otherwise. We now do the same for the tree properties. The conditions of the definition below are inequalities with a gap, which is what makes them survive in elementary extensions of the structure $\mathcal{S}_\varphi$.

Definition 5.2. Let Y be a set and $\mathcal{R}$ a family of real functions on Y.

(1) For an integer $k \geq 2$, a sequence $(r_j)_{j\in\mathbb{N}}$ in $\mathcal{R}$ is *uniformly k-inconsistent* if there are reals $\alpha < \beta$ such that each r_j takes a value $\geq \beta$ somewhere on Y, and for all $j_1 < \cdots < j_k$ no $y \in Y$ satisfies $r_{j_\ell}(y) > \alpha$ for $\ell = 1, \ldots, k$.

(2) The family $\mathcal{R}$ has the *tree property of the second kind* (TP_2) if there are $k \geq 2$, reals $\alpha < \beta$ and an array $(r_{ij})_{i,j\in\mathbb{N}}$ in $\mathcal{R}$ such that every row is uniformly k-inconsistent with the constants α and β, and for every $f\colon \mathbb{N} \to \mathbb{N}$ and every $n \in \mathbb{N}$ there is $y \in Y$ with $r_{if(i)}(y) \geq \beta$ for all $i < n$.

The tree property of Definition 2.1, its variant of the first kind, and TP_2 all require infinite sequences of instances that are k-inconsistent, for some integer $k \geq 2$: the children of a node, or a row. When these notions are transported to a family of functions as in Definition 5.2, those sequences become uniformly k-inconsistent. A family that, for no integer $k \geq 2$, contains a uniformly k-inconsistent sequence therefore has none of the three tree properties. Applied to the family of responses of an observable, condition (2) of Definition 5.2 says that actual preparations witness the tree property of the second kind of the atomic formula, in the real-valued logic of the second paper; the gap between the constants α and β of Definition 5.2 is what the definitions of dividing in real-valued logic require [BY03].

5.4. **The Pauli observable.** The *Pauli observable* is the diagonal observable of the symplectic form ω on $V \times V$, where V is the space of labels of the Pauli strings on countably many qubits that act nontrivially on only finitely many of them (see Section 3): Alice and Bob each hold a Pauli label, and the observable reports whether the two Pauli strings anticommute. We call Alice's label the check and Bob's label the error; the observable then reports one syndrome bit. For a mixture μ, the response at an error e is the probability that a check drawn with law μ anticommutes with e. By Lemma 3.2, which provides an error with any prescribed syndrome with respect to finitely many linearly independent checks, the observable has the independence property already on pure preparations.

Lemma 5.3. *For the Pauli observable, the space K_ω is the group V^* of all linear functionals $V \to \mathbb{F}_2$.*

Proof. Each $\omega(\cdot, e)$ is linear, and the linear functionals form a closed subset of $\{0,1\}^V$. Conversely, a basic neighborhood of a linear functional λ prescribes the values of λ on finitely many vectors, which may be taken linearly independent, and Lemma 3.2 gives e with $\omega(\cdot, e)$ in the neighborhood. □

The functionals of V^* have a direct physical meaning. Let $\hat{V} = \prod_{k\in\mathbb{N}} \mathbb{F}_2^2$, with the product topology, be the group of labels of Pauli strings of arbitrary support, the support of a string

being the set of qubits on which it acts nontrivially; it contains V as the strings of finite support. For $a \in V$ and $u \in \hat{V}$, the value $\omega(a,u)$ is defined by the formula of Section 3, which is a finite sum because a has finite support. For p with $0 < p \le 1$, let ν_p be the *depolarizing noise* of rate p: the product measure on $\hat{V}$ under which the labels on distinct qubits are independent, each equal to $(0,0)$ with probability $1-p$ and to each of $(1,0)$, $(0,1)$ and $(1,1)$ with probability $p/3$. A *Pauli channel* on countably many qubits is a Radon probability measure on $\hat{V}$, the law of a random error applied to the qubits.

Proposition 5.4. *Let $\hat{V}$ be the group of labels of Pauli strings of arbitrary support, and for $0 < p \le 1$ let ν_p be the depolarizing noise of rate p. Then the following hold:*

(a) *the map $u \mapsto \omega(\cdot, u)$ is an isomorphism of topological groups from $\hat{V}$ onto V^*, which carries V onto the functionals of the actual errors;*

(b) *the probes of the Pauli observable are the Pauli channels on countably many qubits, and the response of a mixture μ at a Pauli channel ν is the probability that a check drawn with law μ anticommutes with an independent error drawn with law ν;*

(c) *the depolarizing noise ν_p is concentrated on errors of infinite support, and for a label $a \in V$ that acts nontrivially on w qubits we have*

$$r_{\delta_a}(\nu_p) = \tfrac{1}{2}\big(1 - (1 - \tfrac{4p}{3})^w\big).$$

Proof. (a) The map is a homomorphism. Write X_k and Z_k for the labels of the strings X and Z on the qubit k. Then the values $\omega(X_k,u)$ and $\omega(Z_k,u)$ are the two coordinates of u at the qubit k, so the map is injective, and every $\lambda \in V^*$ is the image of the vector $u = (x,z)$ with $z_k = \lambda(X_k)$ and $x_k = \lambda(Z_k)$ for every k, because λ and $\omega(\cdot,u)$ are linear and agree on the basis $\{X_k, Z_k : k \in \mathbb{N}\}$ of V. The value $\omega(a,u)$ depends on finitely many coordinates of u, so the map is continuous, and a continuous bijection between compact Hausdorff spaces is a homeomorphism.

(b) By Lemma 5.1, Lemma 5.3 and (a), the probes are the Radon probability measures on $\hat{V}$, and $r_\mu(\nu) = \int \sum_a \mu(a)\,\omega(a,u)\,d\nu(u)$, which is the probability stated, by Fubini's theorem.

(c) Under ν_p, the events "the error acts nontrivially on the qubit k" are independent and have probability p, so, by the Borel–Cantelli lemma, almost every error acts nontrivially on infinitely many qubits. At each of the w qubits where the string with label a acts nontrivially, its factor anticommutes with two of the three nontrivial Pauli matrices, hence with the factor of the error with probability $2p/3$, independently of the other qubits. The strings anticommute exactly when an odd number of these w events occur, and this has probability $\frac{1}{2}\big(1 - (1 - \frac{4p}{3})^w\big)$. □

In what follows we identify V^* with $\hat{V}$ through Proposition 5.4(a) and call its elements *idealized errors*. By Proposition 5.4(c), the depolarizing noise ν_p of each rate p, which models the noise of a memory on countably many qubits, is a probe of the Pauli observable concentrated on idealized errors that are not actual: with probability one, an error drawn from ν_p acts on infinitely many qubits. Alice's checks see such an error through finitely many qubits at a time, and the response of a check of weight w, one that acts nontrivially on w qubits, tends to $\frac{1}{2}$, the value of a fair coin, as w grows. For $a \in V$, let χ_a be the character of V^* given by $\chi_a(\lambda) = (-1)^{\lambda(a)}$, a continuous homomorphism of V^* into the multiplicative group $\{\pm 1\}$. Alice's pure responses are the characters of this compact group up to an affine change: the pure response r_{δ_a} of a label a takes the value $\lambda(a) = (1 - \chi_a(\lambda))/2$ at an idealized error λ.

5.5. **The Rado observable.** The *Rado observable* is the diagonal observable of the edge relation E on $\Gamma \times \Gamma$, where (Γ, E) is the Rado graph of Section 3: Alice and Bob each hold a

site of a random network of couplings, and the observable reports whether the two sites are coupled. By the extension property of the Rado graph, which provides a vertex adjacent to every vertex of a finite set and to none of another finite set disjoint from it, the observable has the independence property already on pure preparations. We say that the diagonal observable C_φ of a function $\varphi\colon A\times B\to\{0,1\}$ has *freely independent* pure responses when $K_\varphi=\{0,1\}^A$, that is, when every pattern of answers to all of Alice's labels is an idealized label of Bob.

Lemma 5.5. *For the Rado observable, the space K_E is the whole Cantor cube $\{0,1\}^\Gamma$. Hence the Rado observable has freely independent pure responses, while the Pauli observable does not.*

Proof. A basic open subset of $\{0,1\}^\Gamma$ prescribes adjacency to the vertices of a finite set Γ_1 and nonadjacency to those of a finite set Γ_0, and the extension property gives a vertex outside $\Gamma_1\cup\Gamma_0$ with that pattern. By Lemma 5.3, the space K_ω is the group V^*, which is a proper subset of $\{0,1\}^V$. □

The pure probes of the Rado observable are arbitrary sets of neighbors, that is, arbitrary subsets of Γ, and its probes are random sets of neighbors. The response of a mixture μ at a probe ν is the probability that a site drawn with law μ lies in a set of neighbors drawn, independently, with law ν; it depends on ν only through the marginals $\nu\{\lambda:\lambda(a)=1\}$, for $a\in\Gamma$, the probabilities that the individual sites are neighbors. The Pauli observable sees more of a probe: its pure responses are, up to an affine change, the characters of the compact group of idealized errors, and products of characters are again characters. In Sections 7 and 8 we show what this difference does to the tree property.

6. The theory of mixed preparations has the tree property of the second kind

This section is devoted to Theorem 1.1(C), and its idea is a comparison of two averages. Consider a finite set of Alice's labels and a half of it. The difference between the response to the uniform mixture on the half and the response to the uniform mixture on the complementary half reaches its maximum, 1, at a probe exactly when the probe answers 1 on the first half and 0 on the second. The differences given by two halves that meet in a quarter of the set cannot both be nearly maximal, while the independence property of the pure responses supplies a pure probe with any prescribed answers on finitely many disjoint finite sets of labels. These three facts, established in the proof of Theorem 6.1, together give the tree.

Theorem 6.1 (Mixed preparations and the tree property of the second kind). *Let C_φ be the diagonal observable of a function $\varphi\colon A\times B\to\{0,1\}$ on countable sets A and B, and assume that there are distinct labels $a_0,a_1,\dots$ in A such that for all finite disjoint $N_1,N_0\subseteq\mathbb{N}$ there is a label $b\in B$ with $\varphi(a_n,b)=1$ for $n\in N_1$ and $\varphi(a_n,b)=0$ for $n\in N_0$. Then the affine formula*

$$\psi(x;y_1,y_2)=R(y_1,x)-R(y_2,x),$$

in which the predicate R is the response, the variable x is of the sort of probes and the variables y_1,y_2 are of the sort of preparations, has the tree property of the second kind in the real-valued theory of the structure $\mathcal{S}_\varphi$ of Alice's mixtures and Bob's probes, with $k=2$, $\alpha=1/2$ and $\beta=1$. Here k, α and β are the constants of Definition 5.2, applied to the family of instances of ψ, as functions on the probes, in an $\aleph_1$-saturated elementary extension of $\mathcal{S}_\varphi$. Moreover, every simple diagonal observable is NIP.

Proof. We first build finite arrays in $\mathcal{S}_\varphi$ itself and then pass to an elementary extension by compactness.

Let $N \geq 1$ be an integer, and choose an integer t with $2^t > N$. Partition $2^t N$ of the labels a_n into N blocks $J_0, \ldots, J_{N-1}$ of size 2^t, and identify each block with $\mathbb{F}_2^t$. Choose distinct nonzero vectors $w_0, \ldots, w_{N-1}$ in $\mathbb{F}_2^t$, and for $i, j < N$ let $H_{ij} \subseteq J_i$ be the set of $v \in \mathbb{F}_2^t$ with $w_j \cdot v = 1$, where the dot product $w_j \cdot v$ is computed in $\mathbb{F}_2$. Each H_{ij} has 2^{t-1} elements, and for $j \neq j'$ the set $H_{ij} \cap H_{ij'}$ has 2^{t-2} elements, since w_j and $w_{j'}$ are linearly independent. Let μ_{ij}^+ and μ_{ij}^- be the uniform probability measures on H_{ij} and on $J_i \setminus H_{ij}$.

For a probe ν, write $\pi_a(\nu) = \nu\{\lambda : \lambda(a) = 1\} \in [0,1]$ for its marginal at the label a; by Lemma 5.1, the response is then $R(\mu, \nu) = \sum_a \mu(a)\pi_a(\nu)$. With $m = 2^t$, we have

$$\psi(\nu; \mu_{ij}^+, \mu_{ij}^-) = \frac{2}{m}\Big(\sum_{a \in H_{ij}} \pi_a(\nu) - \sum_{a \in J_i \setminus H_{ij}} \pi_a(\nu) \Big) \leq 1.$$

For $j \neq j'$ in the same row, adding the two values gives

$$\frac{4}{m}\Big(\sum_{a \in H_{ij} \cap H_{ij'}} \pi_a(\nu) - \sum_{a \in J_i \setminus (H_{ij} \cup H_{ij'})} \pi_a(\nu) \Big) \leq \frac{4}{m} \cdot \frac{m}{4} = 1,$$

so the two values are never both greater than $1/2$. For every function f from $\{0, \ldots, N-1\}$ to itself, the hypothesis applied to the finite disjoint sets $\bigcup_i H_{if(i)}$ and $\bigcup_i (J_i \setminus H_{if(i)})$ gives a label b, and at the pure probe of $\varphi(\cdot, b)$ every value $\psi(\cdot; \mu_{if(i)}^+, \mu_{if(i)}^-)$ with $i < N$ equals 1.

Thus, for every N, the structure $\mathcal{S}_\varphi$ satisfies the following conditions on $2N^2$ preparations $y_{ij}^\pm$, for $i, j < N$: for every function f from $\{0, \ldots, N-1\}$ to itself, the value of $\sup_x \min_{i<N} \psi(x; y_{if(i)}^+, y_{if(i)}^-)$ is 1; and for all $i < N$ and all distinct $j, j' < N$, the value of $\sup_x \min\big(\psi(x; y_{ij}^+, y_{ij}^-), \psi(x; y_{ij'}^+, y_{ij'}^-)\big)$ is at most $1/2$. These are finitely many closed conditions for each N, and the conditions for N are inherited by subarrays. By compactness, an $\aleph_1$-saturated elementary extension of $\mathcal{S}_\varphi$ contains an array $(y_{ij}^\pm)_{i,j \in \mathbb{N}}$ satisfying them for all N; by saturation, the suprema along paths are attained. This is the tree property of the second kind, which is incompatible with simplicity in real-valued logic [BY03].

It remains to be shown that every simple diagonal observable is NIP. The hypothesis of the theorem is the independence property of the pure responses, in the form used in the second paper: actual labels are dense in K_φ, and a finite pattern of answers to the labels a_n prescribes a clopen subset of K_φ, so the hypothesis says exactly that the functions $\lambda \mapsto \lambda(a_n)$ form an independent sequence on K_φ, in the sense recalled before Definition 5.2. For a diagonal observable, the pure and the mixed responses have the independence property together, since a mixed response is an average of pure ones. Indeed, suppose that the pure responses have no independent sequence. By Rosenthal's dichotomy, in its original combinatorial form, every bounded sequence of real functions on a set has a subsequence that either converges pointwise or is independent; hence every sequence of pure responses has a subsequence that converges pointwise on K_φ, and this subsequence is weakly Cauchy in $C(K_\varphi)$ by dominated convergence: its values under every continuous linear functional form a Cauchy sequence. By Stegall's theorem, as in the second paper, this property passes to the absolutely convex hull of the pure responses, the set of their finite combinations with coefficients of total absolute value at most 1, and, by approximation, to the norm closure of that hull, which contains the mixed responses. Evaluation at a probe is a continuous linear functional on $C(K_\varphi)$, and every sequence of mixed responses therefore has a subsequence that converges pointwise on the probe space, which no independent sequence has. Consequently, if a diagonal observable

has the independence property, the hypothesis holds, and the real-valued theory of $\mathcal{S}_\varphi$ has the tree property of the second kind, so the observable is not simple. □

The array that witnesses the tree property of the second kind in the proof of Theorem 6.1 lies in an $\aleph_1$-saturated elementary extension of $\mathcal{S}_\varphi$. In the next two sections we show that, for the Pauli observable, actual preparations already witness the tree property of the second kind, for the atomic formula, the response itself, and that, for the Rado observable, every array that witnesses it for a quantifier-free formula contains idealized preparations.

7. The Pauli observable: syndrome sectors and the first check to fire

We turn to Theorem 1.1(D): for the Pauli observable, actual preparations witness the tree property of the second kind. The tool is the expression of the projector onto the states of a given syndrome as an average over the group generated by the checks; we first state it for responses, as a formula for the indicator of a syndrome sector, the set of idealized errors with a given syndrome. The idea of the theorem is then to let a single mixture ask a question about many checks at once. Given a block, a sequence of linearly independent checks, let Alice draw a Pauli string uniformly from the coset of one of its checks modulo the span of the checks before it. In the proof of Theorem 7.2 we show that if the earlier checks are silent on Bob's error, that is, commute with it, and the chosen check fires, that is, anticommutes with it, then every string in the coset anticommutes with the error, and that if one of the earlier checks fires, exactly half of the strings do. Two different checks of a block cannot both be the first to fire, and the syndromes with respect to several blocks can be prescribed separately when all their checks together are linearly independent (see Lemma 3.2).

7.1. **Syndrome sectors.** Let d be a positive integer, let $g_1, \dots, g_d$ be linearly independent checks in the space V of labels of Pauli strings of finite support, with span G, and let a syndrome $\sigma \in \mathbb{F}_2^d$ be given. We regard σ as the linear functional on G with $\sigma(g_i) = \sigma_i$, and we call the set $\Sigma_\sigma = \{\lambda \in V^* : \lambda(g_i) = \sigma_i \text{ for } i \le d\}$ of idealized errors with syndrome σ its *syndrome sector*. For $s \in G$ we write r_s for the pure response r_{δ_s}, the response of the point mass at s, and we recall that χ_s is the character of V^* given by $\chi_s(\lambda) = (-1)^{\lambda(s)}$.

Proposition 7.1. *Let d be a positive integer, let $g_1, \dots, g_d \in V$ be linearly independent checks with span G, and let $\sigma \in \mathbb{F}_2^d$ be a syndrome, regarded as the linear functional on G with $\sigma(g_i) = \sigma_i$ for $i \le d$. Then the indicator of the syndrome sector Σ_σ satisfies, at every pure probe $\lambda \in V^*$,*

$$\mathbf{1}_{\Sigma_\sigma}(\lambda) = 2^{-d} \sum_{s \in G} (-1)^{\sigma(s)} \chi_s(\lambda) = 2^{-d} \sum_{s \in G} (-1)^{\sigma(s)} - 2^{1-d} \sum_{s \in G} (-1)^{\sigma(s)} r_s(\lambda).$$

Consequently, if $\sigma \neq 0$, then $\mathbf{1}_{\Sigma_\sigma} = r_{\mu_\sigma^1} - r_{\mu_\sigma^0}$ on pure probes, where μ_σ^1 and μ_σ^0 are the uniform probability measures on $\{s \in G : \sigma(s) = 1\}$ and on $\{s \in G : \sigma(s) = 0\}$; and $\mathbf{1}_{\Sigma_0} = 1 - 2r_{\mu_G}$, where μ_G is the uniform probability measure on G. In particular, for every probe ν, the probability that an error drawn from the Pauli channel ν has syndrome σ is $r_{\mu_\sigma^1}(\nu) - r_{\mu_\sigma^0}(\nu)$ if $\sigma \neq 0$, and $1 - 2r_{\mu_G}(\nu)$ if $\sigma = 0$.

Proof. The map $s \mapsto (-1)^{\sigma(s)} \chi_s(\lambda) = (-1)^{(\sigma+\lambda)(s)}$ is a character of the finite group G; its sum over G is 2^d if the linear functional $\sigma + \lambda$ vanishes on G, that is, if $\lambda \in \Sigma_\sigma$, and 0 otherwise, since a nonzero functional on G takes the value 1 on exactly half of G. This is the first equality, and the second follows from $\chi_s = 1 - 2r_s$. If $\sigma \neq 0$, the first sum on the right vanishes, and the two level sets of σ have 2^{d-1} elements each; grouping the second sum by the value of σ gives $r_{\mu_\sigma^1} - r_{\mu_\sigma^0}$. If $\sigma = 0$, the first sum is 2^d, and the second is $2^d r_{\mu_G}$.

Both sides of each identity are affine in the probe, and integration against ν, together with Proposition 5.4(b), yields the last statement. □

When the checks commute pairwise and carry signs, the operator $2^{-d}\sum_{s\in G}(-1)^{\sigma(s)}g_s$, where g_s is the element with label s of the group generated by the signed checks, is the projector onto the states of syndrome σ [Got97], and the probability of the outcome σ when the checks are measured on a code state corrupted by a Pauli channel ν is given by Proposition 7.1. The proposition requires no commutation: the checks need only be linearly independent. The coefficients of the indicator, as an affine combination of Alice's responses, have total absolute value at most $1+2=3$, whatever the number d of checks.

7.2. **The first check to fire.** The next theorem arranges the syndrome sectors of the initial segments of blocks of checks in an array that witnesses the tree property of the second kind.

Theorem 7.2 (The first check to fire). *Let $(g_{im})_{i,m\in\mathbb{N}}$ be linearly independent vectors of V, put $W_{ij}=\operatorname{span}(g_{im}:m<j)$ for $i,j\in\mathbb{N}$, and let μ_{ij} be the uniform probability measure on the coset $g_{ij}+W_{ij}$. For the Pauli observable, the following hold:*

(a) *for every $f\colon\mathbb{N}\to\mathbb{N}$ and every $n\in\mathbb{N}$, there is an error $e\in V$ with $r_{\mu_{if(i)}}(e)=1$ for all $i<n$, and there is an idealized error $\lambda\in V^*$ with $r_{\mu_{if(i)}}(\lambda)=1$ for all i;*

(b) *for every i, every $j\neq j'$ and every probe $\nu\in\mathcal{P}(V^*)$, we have $\min\bigl(r_{\mu_{ij}}(\nu),r_{\mu_{ij'}}(\nu)\bigr)\le 2/3$, and the bound $2/3$ is attained.*

Consequently, the family of responses of Alice has the tree property of the second kind in the sense of Definition 5.2, with $k=2$, $\alpha=2/3$ and $\beta=1$.

Proof. We first compute the response of μ_{ij} at a pure probe $\lambda\in V^*$. If λ does not vanish on W_{ij}, then the functional λ takes the value 1 on exactly half of the coset $g_{ij}+W_{ij}$, since a nonzero functional on a finite $\mathbb{F}_2$-space is balanced, and translation by g_{ij} either preserves or exchanges its level sets. If λ vanishes on W_{ij}, then the functional λ is constant on the coset, with value $\lambda(g_{ij})$. Hence

$$r_{\mu_{ij}}(\lambda)=\begin{cases}1 & \text{if } \lambda(g_{im})=0 \text{ for } m<j \text{ and } \lambda(g_{ij})=1,\\ 1/2 & \text{if } \lambda(g_{im})=1 \text{ for some } m<j,\\ 0 & \text{otherwise.}\end{cases}\tag{1}$$

In words, the response is 1 exactly when g_{ij} is the first check of the block $(g_{im})_{m\in\mathbb{N}}$, which we call block i, to fire, that is, to take the value 1 under λ.

(a) Let f and n be given. The conditions $\lambda(g_{im})=0$ for $m<f(i)$ and $\lambda(g_{if(i)})=1$, for $i<n$, concern finitely many linearly independent vectors, so Lemma 3.2 gives an error e with $\omega(\cdot,e)$ satisfying them, and by (1) the response $r_{\mu_{if(i)}}(e)$ equals 1 for all $i<n$. The conditions for all i together define a nonempty closed subset of V^*, since the vectors g_{im} are linearly independent; any λ in it works.

(b) By symmetry, we may assume that $j<j'$. By (1), the pair $\bigl(r_{\mu_{ij}}(\lambda),r_{\mu_{ij'}}(\lambda)\bigr)$ at a pure probe takes one of the values

$$(1,\tfrac12),\quad(\tfrac12,\tfrac12),\quad(0,\tfrac12),\quad(0,1),\quad(0,0).$$

Indeed, if the first check of block i to fire among $g_{i0},\dots,g_{ij'}$ has index m, the pair is $(\frac12,\frac12)$ for $m<j$, $(1,\frac12)$ for $m=j$, $(0,\frac12)$ for $j<m<j'$ and $(0,1)$ for $m=j'$; if none fires, the pair is $(0,0)$. Responses are affine in the probe, so at a mixed probe ν the pair lies in the convex hull of these five points. On that hull the minimum of the two coordinates is at most $2/3$, with

equality only at the point $\frac{2}{3}(1,\frac{1}{2})+\frac{1}{3}(0,1)=(\frac{2}{3},\frac{2}{3})$. This point is the pair of responses at the probe $\frac{2}{3}\delta_{\lambda_1}+\frac{1}{3}\delta_{\lambda_2}$, where λ_1 and λ_2 are idealized errors under which g_{ij} and $g_{ij'}$, respectively, are the first checks of block i to fire; such idealized errors exist by (a).

Statements (a) and (b) are the two clauses of Definition 5.2 for the array $(r_{\mu_{ij}})$, with $k=2$, $\alpha=2/3$ and $\beta=1$. □

Remark 7.3. For qudits of prime dimension p, the quantum systems with Hilbert space $\mathbb{C}^p$, the labels form a symplectic space over the field $\mathbb{F}_p$ with p elements, and the response of a label at an error is 1 when their commutation phase, the pth root of unity by which the products of the two strings in the two orders differ, is nontrivial and 0 otherwise. The proof of Theorem 7.2 adapts as follows. A nonzero functional on a finite $\mathbb{F}_p$-space takes each value equally often, so in (1) the conditions $\lambda(g_{ij})=1$ and $\lambda(g_{im})=1$ become $\lambda(g_{ij})\neq 0$ and $\lambda(g_{im})\neq 0$, and the response $\frac{1}{2}$ becomes $\vartheta=\frac{p-1}{p}$. For $j<j'$, the pair of responses at a pure probe takes one of the values $(1,\vartheta)$, (ϑ,ϑ), $(0,\vartheta)$, $(0,1)$ and $(0,0)$. The point (ϑ,ϑ) lies below the segment from $(1,\vartheta)$ to $(0,1)$, since $\vartheta<1-\vartheta+\vartheta^2$, and on that segment the two coordinates are equal at the point with first coordinate $\frac{p}{p+1}$. Hence the bound $\frac{2}{3}$ of Theorem 7.2(b) becomes $\frac{p}{p+1}$, and it is attained.

In the notation of Theorem 7.2 and the language of Proposition 7.1, the event "the check g_{ij} fires first in block i" is the syndrome sector of the checks $g_{i0},\dots,g_{ij}$ with syndrome $(0,\dots,0,1)$, whose two level sets are the coset $g_{ij}+W_{ij}$ and the group W_{ij}. Hence its indicator is $r_{\mu_{ij}}-r_{\mu^0_{ij}}$ on pure probes, where μ^0_{ij} is the uniform probability measure on W_{ij}. Only the linear independence of the checks enters Theorem 7.2; when the checks commute pairwise, each W_{ij} is the set of labels of a stabilizer group, and μ_{ij} draws uniformly from a coset of it. The simplest instance is the repetition code.

Example 7.4. Index the qubits by pairs $(i,m)\in\mathbb{N}^2$, and for each $i\in\mathbb{N}$ call the qubits (i,m), $m\in\mathbb{N}$, the chain i; let $g_{im}=Z_{i,m}Z_{i,m+1}$ be the check of the bit-flip repetition code that compares the neighboring qubits m and $m+1$ of the chain i, where $X_{i,m}$ and $Z_{i,m}$ denote the bit flip and the phase flip on the qubit (i,m). These checks commute pairwise and are linearly independent. For a bit flip $e=\prod_{i,m}X_{i,m}^{x_{i,m}}$, with exponents $x_{i,m}\in\mathbb{F}_2$, we have $\omega(g_{im},e)=x_{i,m}+x_{i,m+1}$, which is 1 if and only if exactly one of the qubits m and $m+1$ of the chain i is flipped, that is, if and only if e has a domain wall between them. By (1), the response at e of the uniform probability measure μ_{ij} on the coset of g_{ij} modulo the span of $g_{i0},\dots,g_{i,j-1}$, as in Theorem 7.2, is 1 exactly when the first domain wall of e on the chain i lies between the qubits j and $j+1$. For this code, Theorem 7.2 reads as follows: the chains are independent code blocks of a memory, an error cannot have two first domain walls on the same chain, and the positions of the first domain walls on finitely many chains can be prescribed independently by a single error. The bit flips on the chain i with zero syndrome are the identity and the flip of every qubit of the chain, which acts on infinitely many qubits: the logical bit flip of an infinite chain is an idealized error, an element of $\hat{V}\setminus V$.

Remark 7.5. The mechanism behind Theorem 7.2 is Fourier analysis on the compact group V^*: since the characters of V^* separate points and are closed under multiplication, the linear span of Alice's responses and the constants is dense in $C(V^*)$ by the Stone–Weierstrass theorem; it is an algebra, because $2r_ar_b=r_a+r_b-r_{a+b}$ for the pure responses $r_a=r_{\delta_a}$, $a\in V$. By Proposition 7.1, the indicators of syndrome sectors, which are functions of supremum norm one, lie in this span with coefficients of total absolute value at most 3. In the first-order structure (V,ω), by contrast, a formula speaks about a bounded number of labels, and, by Theorem 3.3, no formula has the tree property.

By Theorem 6.1, the real-valued theory of the structure $\mathcal{S}_\omega$ of mixed preparations and probes of the Pauli observable is not simple, although the first-order theory of (V,ω) is; in Theorem 7.2 the tree property of the second kind is witnessed by actual preparations, for the atomic formula, the response itself, rather than for the difference of two responses.

8. The Rado observable: marginals and idealized preparations

In this section we prove Theorem 1.1(E), which concerns the Rado observable. By Theorem 6.1, the real-valued theory of the structure of its mixed preparations and probes has the tree property of the second kind, as that of the Pauli observable does. In Theorem 8.1, Corollary 8.2 and Theorem 8.3 we show that every array that witnesses this property for a quantifier-free formula contains idealized preparations, because the conditions that actual mixtures impose on the marginals of a probe, the probabilities with which it declares the individual vertices neighbors, are compatible on disjoint finite sets of vertices. After Corollary 8.2 we note that the same holds for the random bipartite graph, the Fraïssé limit of the finite bipartite graphs with the two sides as two sorts, which has exactly the shape of a bipartite observable: Alice's labels form one side, and Bob's the other.

8.1. **Affine combinations of responses.** Consider a diagonal observable with freely independent pure responses, that is, one for which every pattern $\lambda \in \{0,1\}^A$ of answers to Alice's labels, the elements of A, is an idealized label of Bob. The response of a mixture depends on a probe ν only through its marginals $\pi_a(\nu) = \nu\{\lambda : \lambda(a) = 1\}$, for $a \in A$, as for the Rado observable in Section 5.5, and every vector of marginals in $[0,1]^A$ is attained by a product measure. A bounded affine combination of responses is therefore a function

$$h_{c,\gamma}(\nu) = \gamma + \sum_{a\in A} c(a)\,\pi_a(\nu), \qquad \gamma \in \mathbb{R},\ c \in \ell^1(A),$$

and its supremum over the probe space is $\gamma + \sum_a c(a)^+$, attained at the pure probe $\lambda = \mathbf{1}_{\{c>0\}}$. Two such functions can conflict only through the labels on which both put weight. In the proof of Theorem 8.1 below we show, by a gliding hump, that a sequence of such functions along which $|\gamma| + \|c\|_1$ stays bounded has a subsequence whose members put their weight, up to small errors, on a common finite set of labels and on pairwise disjoint finite blocks; for every positive integer k, a single pure probe then nearly maximizes k of them.

Theorem 8.1 (Compatibility of affine conditions). *Let C_φ be the diagonal observable of a function $\varphi\colon A \times B \to \{0,1\}$ on countable sets A and B, with freely independent pure responses, and let $\kappa > 0$ be given. Let $\mathcal{R}_\kappa$ be the family of functions $\gamma + \sum_{i<n} c_i r_{\mu_i}$ on the probe space $\mathcal{P}(\{0,1\}^A)$ of Radon probability measures on $\{0,1\}^A$, where $n \in \mathbb{N}$, the μ_i are mixtures of Alice, and γ and the c_i are reals with $|\gamma| + \sum_i |c_i| \le \kappa$. Then the family $\mathcal{R}_\kappa$ contains no uniformly k-inconsistent sequence, for any integer $k \ge 2$. In particular, no tree property is witnessed in $\mathcal{R}_\kappa$.*

Proof. Let $(h_j)_{j\in\mathbb{N}}$ be a sequence in $\mathcal{R}_\kappa$ and let $\alpha < \beta$ be such that each h_j takes a value $\ge \beta$. We show that for every integer $k \ge 2$ there are $j_1 < \cdots < j_k$ and a pure probe λ with $h_{j_\ell}(\delta_\lambda) > \alpha$ for $\ell = 1,\dots,k$.

Write $h_j = \gamma_j + \sum_i c_{j,i} r_{\mu_{j,i}}$, with real coefficients $c_{j,i}$ and mixtures $\mu_{j,i}$ of Alice; then $h_j = h_{c_j,\gamma_j}$ with $c_j = \sum_i c_{j,i}\mu_{j,i}$, whence $\|c_j\|_1 \le \kappa$. Since A is countable, a diagonal argument gives a subsequence, still denoted (c_j), that converges pointwise on A to some c; by Fatou's lemma, the limit satisfies $\|c\|_1 \le \kappa$. Put $\eta = (\beta - \alpha)/5$ and choose a finite set

$F \subseteq A$ with $\|c \restriction (A \setminus F)\|_1 < \eta$. Put $F_0 = F$, and choose inductively indices $j_1 < \cdots < j_k$ and finite sets $F_0 \subseteq F_1 \subseteq \cdots \subseteq F_k$ such that

$$\|(c_{j_\ell} - c) \restriction F_{\ell-1}\|_1 < \eta \quad \text{and} \quad \|c_{j_\ell} \restriction (A \setminus F_\ell)\|_1 < \eta;$$

the first condition holds for all large indices by pointwise convergence, and the second for a large enough finite set. Define $\lambda \in \{0,1\}^A$ by $\lambda(a) = 1$ exactly when $a \in F$ and $c(a) > 0$, or $a \in F_\ell \setminus F_{\ell-1}$ and $c_{j_\ell}(a) > 0$ for some $\ell \leq k$.

Let $\ell \leq k$ be given. The deficit of λ for h_{j_ℓ} is

$$\sup h_{j_\ell} - h_{j_\ell}(\delta_\lambda) = \sum_{\lambda(a)=0} c_{j_\ell}(a)^+ + \sum_{\lambda(a)=1} c_{j_\ell}(a)^-,$$

and we bound its contribution on four disjoint pieces of A. On $F_\ell \setminus F_{\ell-1}$ the sign pattern of λ is that of c_{j_ℓ}, and the contribution is 0. On F it is that of c, and a label where the signs of c_{j_ℓ} and c disagree contributes at most $|c_{j_\ell}(a) - c(a)|$; the total is less than η. On $F_{\ell-1} \setminus F$ the contribution is at most $\|c_{j_\ell} \restriction (F_{\ell-1} \setminus F)\|_1 \leq \|(c_{j_\ell} - c) \restriction F_{\ell-1}\|_1 + \|c \restriction (A \setminus F)\|_1 < 2\eta$. On $A \setminus F_\ell$ it is less than η. Hence $h_{j_\ell}(\delta_\lambda) > \sup h_{j_\ell} - 4\eta \geq \beta - 4\eta > \alpha$. □

Corollary 8.2. *Let $n \in \mathbb{N}$ be given, and let γ and c_i, for $i < n$, be reals. For the Rado observable, no tree property of the affine formula $\gamma + \sum_{i<n} c_i R(y_i, x)$, in which R is the response predicate, the variables y_i are of the sort of preparations and the variable x is of the sort of probes, is witnessed by actual preparations, although the real-valued theory of the structure $\mathcal{S}_E$ of its mixed preparations and probes has the tree property of the second kind.*

Proof. By Lemma 5.5, the Rado observable has freely independent pure responses. With actual preparations as parameters, the instances of the formula lie in the family $\mathcal{R}_\kappa$ of Theorem 8.1 for every $\kappa > 0$ with $\kappa \geq |\gamma| + \sum_{i<n} |c_i|$, and that theorem yields the first statement. The second statement is Theorem 6.1. □

The finite arrays in the proof of Theorem 6.1 are made of actual preparations, and their rows have any prescribed finite length; by Theorem 8.1, the rows cannot be made infinite in $\mathcal{S}_E$. The infinite rows obtained by compactness in an $\aleph_1$-saturated elementary extension must therefore contain idealized preparations. The same holds for the random bipartite graph, whose observable has freely independent pure responses, since every pattern of adjacency and nonadjacency to finitely many vertices of one side is realized by a vertex of the other.

8.2. **Quantifier-free formulas.** In Theorem 8.1 the conditions on a probe are given by affine combinations of responses. In the real-valued logic of the second paper, a quantifier-free formula in a variable x of the sort of probes, built from the response with parameters of the sort of preparations, is a continuous function of finitely many responses, since the responses take values in $[0,1]$ and the connectives are continuous functions. The next theorem is the analogue of Theorem 8.1 for these formulas. Its proof uses the same gliding hump, with one change: a continuous function of responses is no longer maximized by reading off signs (we show this after the proof), so instead of building the probe from signs we glue maximizers of the individual instances on disjoint blocks of labels.

Theorem 8.3 (Compatibility of quantifier-free conditions). *Let C_φ be the diagonal observable of a function $\varphi\colon A \times B \to \{0,1\}$ on countable sets A and B, with freely independent pure responses, let $n \geq 1$ be an integer, and let $u\colon [0,1]^n \to \mathbb{R}$ be a continuous function. Let $\mathcal{R}_u$ be the family of functions*

$$h(\nu) = u\big(r_{\mu_1}(\nu), \dots, r_{\mu_n}(\nu)\big) \qquad (\nu \in \mathcal{P}(\{0,1\}^A)),$$

where $\mu_1, \dots, \mu_n$ range over the mixtures of Alice. Then the family $\mathcal{R}_u$ contains no uniformly k-inconsistent sequence, for any integer $k \geq 2$. In particular, no tree property of the quantifier-free formula $u(R(y_1,x), \dots, R(y_n,x))$, where R is the response predicate, the variable x is of the sort of probes and the variables $y_1, \dots, y_n$ are of the sort of preparations, is witnessed by actual preparations.

Proof. For $\nu \in \mathcal{P}(\{0,1\}^A)$ write $\pi(\nu) \in [0,1]^A$ for its vector of marginals $(\pi_a(\nu))_{a\in A}$, defined before Theorem 8.1. Every $h \in \mathcal{R}_u$ factors as $h(\nu) = H(\pi(\nu))$ with

$$H(q) = u\big(\langle \mu_1, q\rangle, \dots, \langle \mu_n, q\rangle\big), \qquad \langle \mu, q\rangle = \sum_{a\in A} \mu(a) q_a,$$

and every $q \in [0,1]^A$ is the vector of marginals of a product measure. It therefore suffices to work on the cube $[0,1]^A$.

Let $(h_j)_{j\in\mathbb{N}}$ be a sequence in $\mathcal{R}_u$, with parameters $\mu_{j,1}, \dots, \mu_{j,n}$ and associated functions H_j on the cube, and let $\alpha < \beta$ be such that each h_j takes a value $\geq \beta$. We show that for every integer $k \geq 2$ there are $j_1 < \cdots < j_k$ and $q \in [0,1]^A$ with $H_{j_\ell}(q) > \alpha$ for $\ell = 1, \dots, k$. Since u is uniformly continuous, there is $\theta > 0$ such that $|u(s) - u(t)| < \beta - \alpha$ whenever $\max_i |s_i - t_i| < \theta$. For each j choose $q^{(j)} \in [0,1]^A$ with $H_j(q^{(j)}) \geq \beta$. The set A is countable and the cube is compact and metrizable, so a diagonal argument provides a subsequence, still indexed by j, along which $\mu_{j,i} \to \mu_i$ pointwise on A for each $i \leq n$ and $q^{(j)} \to q^\infty$ in $[0,1]^A$. By Fatou's lemma, each limit μ_i has total mass at most 1. Put $\eta = \theta/5$ and choose a finite set $F \subseteq A$ with $\mu_i(A \setminus F) < \eta$ for all $i \leq n$.

Put $F_0 = F$, and choose inductively indices $j_1 < \cdots < j_k$ and finite sets $F_0 \subseteq F_1 \subseteq \cdots \subseteq F_k$ such that, for all $i \leq n$,

$$\sum_{a\in F_{\ell-1}} |\mu_{j_\ell,i}(a) - \mu_i(a)| < \eta, \qquad \sum_{a\in F} |q_a^{(j_\ell)} - q_a^\infty| < \eta, \qquad \mu_{j_\ell,i}(A \setminus F_\ell) < \eta.$$

The first two conditions hold for all large indices, by convergence on the finite sets $F_{\ell-1}$ and F, and the third for a large enough finite set F_ℓ. Define $q \in [0,1]^A$ by $q_a = q_a^\infty$ for $a \in F$, $q_a = q_a^{(j_\ell)}$ for $a \in F_\ell \setminus F_{\ell-1}$ and $\ell \leq k$, and $q_a = 0$ for $a \notin F_k$.

Let $\ell \leq k$ and $i \leq n$ be given. We compare $\langle \mu_{j_\ell,i}, q\rangle$ with $\langle \mu_{j_\ell,i}, q^{(j_\ell)}\rangle$, on the same four pieces of A as in the proof of Theorem 8.1; on each piece the two vectors differ by at most 1 in each coordinate. On $F_\ell \setminus F_{\ell-1}$ they agree. On F the difference is at most $\sum_{a\in F} |q_a^\infty - q_a^{(j_\ell)}| < \eta$, since $\mu_{j_\ell,i}(a) \leq 1$. On $F_{\ell-1} \setminus F$ it is at most $\mu_{j_\ell,i}(F_{\ell-1} \setminus F) \leq \sum_{a\in F_{\ell-1}} |\mu_{j_\ell,i}(a) - \mu_i(a)| + \mu_i(A \setminus F) < 2\eta$. On $A \setminus F_\ell$ it is at most $\mu_{j_\ell,i}(A \setminus F_\ell) < \eta$. Hence $|\langle \mu_{j_\ell,i}, q\rangle - \langle \mu_{j_\ell,i}, q^{(j_\ell)}\rangle| < 4\eta < \theta$ for every $i \leq n$, and the choice of θ gives $H_{j_\ell}(q) > H_{j_\ell}(q^{(j_\ell)}) - (\beta - \alpha) \geq \alpha$. □

For affine combinations, the proof of Theorem 8.1 yields slightly more: there the common witness can be taken pure. For an arbitrary connective u the witness is the product probe whose vector of marginals is the point q constructed in the proof of Theorem 8.3, and it cannot always be taken pure: for $n = 1$, $\mu = \delta_a$ with $a \in A$, and $u(t) = 1 - |2t - 1|$, the instance vanishes at every pure probe and equals 1 at the probes ν with $\pi_a(\nu) = 1/2$.

The difference between the two paradigms lies entirely in the probe space: on the Cantor cube $\{0,1\}^\Gamma$, the space of pure probes of the Rado observable, affine combinations of coordinates see only marginals, and conditions on marginals of disjoint blocks are compatible. On the group V^* of idealized errors of the Pauli observable, the linear relations among the labels make every product of coordinates affine in the coordinates, and they put the indicators of syndrome sectors in the affine span of the responses (see Proposition 7.1).

9. Affine logic and correlated preparations

Finally, we place the results in affine logic, in the form developed by Ben Yaacov, Ibarlucía and Tsankov [BYIT24], whose connectives are the affine maps and in which a mixture of structures is a direct integral. Formula values in a direct integral are integrals of formula values, which is the rule by which a response is computed on a mixed preparation. In this logic the type spaces are compact convex sets, and a structure is *extremal* when it realizes only extreme types, the extreme points of these sets. Ben Yaacov, Ibarlucía and Tsankov [BYIT24] showed that every model of an affine theory is an affine elementary substructure of a direct integral of extremal models. An affine theory is simplicial when its type spaces are Choquet simplices, that is, sets in which every point is the barycenter of exactly one probability measure that is maximal in the Choquet order; when the set is metrizable, the maximal measures are those concentrated on its extreme points. The same authors also showed that every model of a simplicial theory is such a direct integral, uniquely.

Ben Yaacov and Ibarlucía [BYI26] proved that an affine formula stable in almost every member of a measurable field of structures is stable in its direct integral, and that an affine theory is stable exactly when its extremal models are stable. Their definition of stability of a formula in a structure is the double limit condition of the first paper of this series [DI26a]: for every two sequences of elements, one in each variable of the formula, the two iterated limits of its values agree whenever both exist. Mixing in the sense of this paper behaves differently with respect to simplicity: the first-order theories of the Pauli group and of the Rado graph are simple, whereas, by Theorem 6.1, the real-valued theories of their observables have the tree property of the second kind.

The direct integral allows the elements in the two variables of a formula to depend on the same random parameter. For observables this means that Alice and Bob share randomness: they prepare a separable state, a mixture of product states, rather than a single product state. Keisler's randomizations [Kei99, BY13] are of this kind: their elements are random elements of a structure, that is, measurable maps from a common probability space into it. The preparations in Theorem 6.1, by contrast, are product preparations.

10. Questions

Question 10.1. Which diagonal observables with the independence property have the tree property of the second kind witnessed by actual preparations? For the diagonal observable C_φ of a function $\varphi\colon A \times B \to \{0,1\}$ on countable sets A and B, Proposition 7.1, Theorem 7.2 and Theorem 8.1 suggest a criterion in terms of the linear span of its responses in the space $C(K_\varphi)$ of continuous functions on the space K_φ of idealized labels of Bob: whether it contains, with coefficients of bounded total absolute value, the indicators of the sets of an infinite array of clopen sets whose rows consist of pairwise disjoint sets and in which every choice of one set from each of finitely many rows has nonempty intersection.

Question 10.2. By Remark 7.3, the proof of Theorem 7.2 adapts to qudits of prime dimension. For continuous variables, that is, for systems such as a particle on a line, whose position

and momentum take real values, the commutation phase of a translation by a distance x and a momentum kick by $\hbar k$, where $\hbar$ is the reduced Planck constant and the wavenumber k is a real number measured in the inverse of the unit of length, is $e^{\mathrm{i}kx}$, whose real part $\cos kx$ describes an interference fringe. By the second paper, the fringe observable, which asks whether a photon lands on a bright fringe of an interference pattern of wavenumber 8^n, for a label $n \in \mathbb{N}$ of Alice, is not NIP. Where do interference fringes of unbounded wavenumber lie with respect to simplicity?

Question 10.3. In first-order model theory, a simple theory without the independence property is stable. Is an observable stable when its real-valued theory is simple? By Theorem 6.1, every simple diagonal observable is NIP, so for diagonal observables the question is whether those of the second paper that are NIP but unstable, such as the comparison of photon numbers, which asks whether Bob's system has more photons than Alice's, have a tree property once preparations are mixed.

Question 10.4. Does Theorem 6.1 hold for every observable whose responses have the independence property, not only for diagonal ones? The proof uses the values 0 and 1 of pure responses; for general observables they are replaced by two distinct levels, and the averages must be controlled with a gap.

Question 10.5. Is there an analogue of the Kim–Pillay theorem for affine logic, characterizing simple affine theories by an independence relation on extremal types?

Historical remarks

Simple theories were introduced by Shelah [She80] in 1980, together with the tree properties of the first and second kind; the motivating example was the theory of the countable random graph (see [She80, GIL02]). Kim [Kim98] proved in 1998 that forking is symmetric in simple theories, and Kim and Pillay [KP97] proved the independence theorem and the characterization of simplicity by an independence relation in 1997. Ben Yaacov [BY03] extended the theory in 2003 to compact abstract theories, a framework that contains the continuous logic of metric structures. The Rado graph was described by Ackermann [Ack37] in 1937 and by Rado [Rad64] in 1964; by a theorem of Erdős and Rényi [ER63] from 1963, it is the almost sure isomorphism type of the random graph on countably many vertices. Fraïssé [Fra54] introduced his limits in 1954, and Witt's theorem on isometries of forms is in Artin's *Geometric Algebra* [Art57]. Vector spaces of countably infinite dimension over finite fields with a nondegenerate alternating form, such as the Pauli group, are among the Cherlin–Hrushovski smoothly approximable structures [CH03]. Granger [Gra99] studied vector spaces with a generic bilinear form over an arbitrary field in his thesis of 1999 and showed that, over an infinite field carried as a second sort, the theory is not simple; Chernikov and Ramsey [CR16] proved in 2016 that over an algebraically closed field it is NSOP_1: it lacks the 1-strong order property, as every simple theory does. The symplectic description of the Pauli group and of stabilizer codes is due to Gottesman [Got97] and to Calderbank, Rains, Shor and Sloane [CRSS98]; the expression of the projector onto the states of a given syndrome as an average over the stabilizer group, behind Proposition 7.1, is standard in Gottesman's stabilizer formalism. Keisler [Kei99] introduced randomizations in 1999; Ben Yaacov and Keisler [BYK09] treated them as metric structures, and Ben Yaacov [BY13] proved in 2013 that the Keisler randomization of a theory with the independence property has TP_2. Direct integrals and extremal models in affine logic are due to Ben Yaacov, Ibarlucía and Tsankov [BYIT24], and stability in affine logic to Ben Yaacov and Ibarlucía [BYI26]. The first two papers of this series are [DI26a, DI26b].

Declarations

Competing interests. The authors have no competing interests to declare.
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Department of Mathematics, The University of Texas at San Antonio, One UTSA Circle, San Antonio, TX 78249, USA
Email address: Eduardo.Duenez@utsa.edu

Department of Mathematics, The University of Texas at San Antonio, One UTSA Circle, San Antonio, TX 78249, USA
Email address: jose.iovino@utsa.edu